Statistical Treatment Choice with Untested Doses

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Abstract

I extend Manski's (2025) limited-trial dose-choice model to a finite-sample statistical decision setting. A decision-maker chooses from a finite ordered menu of doses to balance efficacy against adverse effects and dose-specific costs, but trial data are observed only for a subset of these doses. Outcome distributions at untested doses are constrained by the assumption that efficacy and adverse effects are weakly increasing in dose, and are hence partially identified. I show that a plug-in rule based on any feasible uniformly mean-consistent estimator is asymptotically minimax-regret optimal. A constrained maximum likelihood estimator satisfies this uniform mean consistency, and the finite-sample suboptimality gap is $O(n_{\min}^{-1/2})$, where $n_{\min}$ is the smallest per-arm sample size in the trial. This plug-in problem can be formulated as a linear program.

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1 Introduction

Whereas clinical trials typically test a placebo and a small number of doses, the clinician or policy-maker often faces a richer menu that includes intermediate or nearby untested doses. The standard empirical approaches are to choose among tested doses only, or to use fitted dose-response curves to evaluate untested doses (Bretz et al., 2005). Both approaches have shortcomings: the first may ignore credible shape information restricting untested doses; the second may extrapolate from tested to untested doses with spurious precision.

This paper takes an intermediate approach. The dose-response curves are assumed to be monotone at the individual potential-outcome level, so that the efficacy and adverse-effects incidence of untested doses can be bounded by those of nearby tested doses. Precise values are not interpolated. A decision-maker (DM) observes trial outcomes for a subset of an ordered dose menu. Outcomes consist of a binary efficacy endpoint and a binary adverse-effect endpoint. Higher dose is assumed to weakly increase both endpoints.

The paper makes three contributions. First, I formalise the finite-sample statistical decision problem induced by limited dose trials under monotone dose response. Second, I prove that feasible plug-in rules are globally asymptotically minimax regret (MMR) optimal whenever the tested-dose distribution estimator is uniformly mean-consistent, and I establish a rate of convergence for a plug-in rule based on a constrained maximum likelihood estimator (MLE). Third, I show that the plug-in problem is a single linear program, and can thus be solved efficiently using standard linear program solvers. Appendix D extends the model and main results to any finite number of endpoints, each with a finite number of ordered values.

Manski (2025) studies the corresponding population problem.[1] If the tested-dose outcome probabilities were known, the DM could evaluate each action by its maximum regret over all monotone completions of the untested-dose outcome probabilities. The DM would then choose the MMR action. The present paper addresses the statistical analogue of this problem: what decision rule should be used when the tested-dose probabilities are themselves estimated from finite-sample trial data?

I study a plug-in, or as-if, analogue of Manski's (2025) minimax-regret action. This rule treats an estimate of the tested-dose outcome probabilities as if it were the true value, and then solves for the minimax-regret action. Following Wald (1950), I use as a benchmark the decision rule that minimises the maximum risk. In this context, risk is the expectation, over data realisations, of the regret loss associated with actions chosen by the rule. I show that the plug-in rule using a uniformly mean-consistent estimator of the tested-dose outcome distribution asymptotically attains minimax risk. That is, for all $\varepsilon > 0$, there exists an $N(\varepsilon) \in \mathbb{N}$ such that if every trial arm has sample size at least $N(\varepsilon)$ then the maximum risk of the plug-in exceeds the minimax risk by less than ε.

The gap between the plug-in rule's worst-case regret risk and the finite-sample minimax value is bounded by a constant times the worst-case expected estimation error. With the

[1] His Appendix 4 also sketches a statistical formulation and plug-in approach.

constrained MLE, this gap is $O(n_{\min}^{-1/2})$. The proof combines a population lower bound on minimax risk with a uniform upper bound on plug-in risk through estimation error.

The linear-program formulation extends Manski's (2025) result for the inner worst-case completion problem to the full population minimax-regret problem, avoiding the outer grid search his method requires.

Related literature. This paper extends Manski's (2025) limited-trial dose-choice model by moving from a population-level action problem to a Wald-style statistical decision problem. A second closely related paper is Higbee (2025). He also studies untested treatments under shape restrictions, considering the finite-sample problem and deriving similar regret bounds and linear programming formulations. His problem differs from that of Manski (2025), and hence from my own, in that he restricts to nonrandomised treatment assignment, considers other shape restrictions in addition to monotonicity, and assumes a welfare criterion that is linear in the outcome. Moreover, his regret bounds are for what I call expected profiled regret loss, whereas I also provide bounds for risk defined by unprofiled regret loss.

My paper also contributes to the wider econometric literature on statistical treatment choice. Stoye (2007) studies population-level minimax-regret treatment choice with incomplete outcome data and finitely many treatments. He assumes that each treatment can attain its upper welfare bound while all other treatments simultaneously attain their lower bounds. Such joint attainability can fail when monotonicity links adjacent untested doses. On the other hand, when no two untested doses are adjacent, the population problem in this paper is a special case of his problem. Manski (2007) and Stoye (2012) study finite-sample minimax-regret choice between two treatments under partial identification arising from missing outcomes or limited experimental validity. Manski (2004) and Stoye (2009) evaluate the regret of empirical success rules in small samples with point identification. Manski and Tetenov (2016) derive sufficient trial sample sizes for near-optimal treatment choice. Montiel Olea et al. (2026) and Yata (2021) study minimax regret when payoffs are partially identified and data are normally distributed. My paper instead uses the multinomial form of the data-generating process.

Hirano and Porter (2009) derive asymptotically optimal statistical treatment rules using limits of experiments. Christensen et al. (2026), Kido (2023), and Xu (2026) analyse decision rules under partial identification using local asymptotic and limits-of-experiments methods to motivate Bayesian, shifted plug-in, and shifted-parameter plug-in rules. My paper does not rely on local asymptotic normality and targets a global rather than local asymptotic minimax criterion.

Manski (2021) discusses the use and evaluation of plug-in, or as-if, decision rules. Plug-in treatment rules are related to empirical welfare maximisation (Kitagawa and Tetenov, 2018), which chooses a treatment policy by maximising the sample analogue of average welfare over a class of feasible policies.

Structure of the Paper. Section 2 uses a numerical example to illustrate that an untested dose can be optimal. Section 3 introduces the general statistical decision problem and the minimax risk benchmark. Section 4 presents the main results. Section 4.1 introduces the plug-in rule and establishes its finite-sample and asymptotic performance guarantees against this benchmark, given a suitable estimator of the identified subparameter. Section 4.2 introduces a constrained maximum likelihood estimator for that subparameter and shows that it has the required properties. Section 4.3 reformulates the plug-in problem as a single linear program. Section 5 illustrates the plug-in rule using data from two studies on the influenza medication oseltamivir.

2 A Motivating Example: Why Untested Doses Matter

In this section, I illustrate Manski's original problem by way of a simple numerical example. I set dose-specific costs to zero, abstract from sampling error and require that the chosen action be nonrandomising. These are expository simplifications, which will be fully relaxed in the general model considered in subsequent sections.

A decision maker (DM) chooses which dose $t \in \{1, 2, 3\}$ of a treatment to give to a patient population, but only observes the outcome probabilities for doses 1 and 3. For both of these tested doses, the DM observes a probability distribution over the four outcomes in Table 1. The table also gives the welfare of each outcome and welfare-minimising and welfare-maximising outcome probabilities for the untested dose $t = 2$.

Outcome			Outcome probabilities			
Cured	Adverse effects	Welfare	$t = 1$	$t = 3$	min $t = 2$	max $t = 2$
No	No	1	0.5	0.3	0.3	0.4
	Yes	0	0.2	0.3	0.4	0.2
Yes	No	6	0.0	0.0	0.0	0.1
	Yes	3	0.3	0.4	0.3	0.3
Expected welfare			1.4	1.5	1.2	1.9

Table 1: Welfare of each outcome and outcome probabilities. The final two columns give welfare-minimising and welfare-maximising outcome probabilities for $t = 2$ that are consistent with monotonicity.

The values for the expected welfare of the tested doses 1 and 3 are 1.4 and 1.5, respectively. Between the tested doses $t = 1$ and $t = 3$, the DM should choose $t = 3$: the increased incidence of adverse effects is outweighed by the increased efficacy of the higher dose. However, the DM also has the opportunity to choose the untested dose $t = 2$. Since efficacy and adverse-effects incidence weakly increase in dose for every patient, we have the

following restrictions for dose $t = 2$:

$$
\begin{aligned}
&0.3 \leqslant \mathbb{P}(\text{not cured}, \text{no adverse effects} \mid t = 2) \leqslant 0.5,\\
&0.3 \leqslant \mathbb{P}(\text{cured}, \text{adverse effects} \mid t = 2) \leqslant 0.4,\\
&0.3 \leqslant \mathbb{P}(\text{cured} \mid t = 2) \leqslant 0.4,\\
&0.5 \leqslant \mathbb{P}(\text{adverse effects} \mid t = 2) \leqslant 0.7.
\end{aligned}
$$

Though these restrictions bound our uncertainty, they do not identify precise values for the outcome probabilities of the untested dose. Two natural criteria for addressing the residual ambiguity are maximin welfare and minimax regret. Maximin welfare maximises welfare in the least favourable state of the world. When dose costs are zero, maximin welfare is extremely ambiguity-averse.[2] The state of the world in which the joint outcome distribution of the untested dose $t = 2$ is exactly equal to that of either tested dose $t \in \{1, 3\}$ is compatible with the monotonicity assumption. It follows that, under the maximin-welfare criterion, both tested doses have weakly higher worst-case welfare than the untested dose. This paper therefore focuses on the minimax regret criterion.

In a given state, the regret of an action is the difference between the maximum welfare attainable across all actions and the welfare yielded by that action. The action's maximum regret is the largest such difference across states. An MMR action minimises this maximum regret. An action with low maximum regret yields welfare uniformly close to the maximum attainable in each state. For a discussion of the merits of minimax regret in prescriptive decision-making, see Manski (2011).

The possible states of the world are the possible values of the four outcome probabilities for the untested dose $t = 2$ that are consistent with monotonicity and the corresponding probabilities for the two tested doses. Since it has already been established that $t = 3$ dominates $t = 1$, the DM can ignore the latter. A regret-maximising state for dose 3 maximises the welfare of dose 2; a regret-maximising state for dose 2 minimises the welfare of the same dose. The respective maximum regret values are therefore $1.9 - 1.5 = 0.4$ for dose 3 and $1.5 - 1.2 = 0.3$ for dose 2. Since the maximum regret of choosing 2 is smaller than that of choosing 3, the minimax-regret dose choice is $t = 2$. In this example, the MMR criterion selected the untested intermediate dose $t = 2$. This shows that choosing only from experimentally tested doses can be strictly suboptimal under an MMR criterion.

3 Setup

Doses and outcomes. A decision maker (DM) chooses the dose $t \in \{1, \ldots, T\}$ of a treatment to give to a population of observationally identical individuals, indexed by $j \in \mathcal{J}$. The decision maker is allowed to randomise, meaning they may choose to assign dose t to some fraction $a_t \in [0, 1]$ of the population. Thus, the space of possible actions, a, is the

[2] The "ultra-pessimism" of maximin welfare is often cited as a shortcoming of the criterion (e.g. Manski, 2004; Montiel Olea et al., 2026; Savage, 1951; Stoye, 2009).

$(T-1)$-simplex,

$$\mathcal{A} := \left\{ a \in [0,1]^T \;\middle|\; \sum_{t=1}^{T} a_t = 1 \right\}.$$

The effects of the treatment are described by two binary endpoints: whether the individual is free from the disease and whether they experience adverse effects. The respective dose-response functions are $y_e^j, y_{\text{ae}}^j : \{1,\dots,T\} \to \{0,1\}$ defined by

$$y_e^j(t) := \mathbf{1}\{\text{individual } j \text{ is free from disease when given dose } t\},$$
$$y_{\text{ae}}^j(t) := \mathbf{1}\{\text{individual } j \text{ experiences adverse effects when given dose } t\}.$$

The binary outcome set is

$$\mathcal{Y} := \{(0,0),(0,1),(1,0),(1,1)\}.$$

Where outcomes appear as subscripts, I use the abbreviations $00, 01, 10, 11$. Since individuals are observationally identical, the dose-response functions $y^j := (y_e^j, y_{\text{ae}}^j)$ are random variables from the perspective of the DM.

In addition, the DM knows ex ante that, at the individual level, both outcomes are monotone increasing in dose. That is:

Assumption 1 (Monotone dose response)**.**

(i) Monotone efficacy: if a particular individual would be disease-free when given dose t, then they would also be disease-free when given any dose $t' \geqslant t$.

(ii) Monotone adverse-effects incidence: if a particular individual would experience adverse effects when given dose t, then they would also experience adverse effects when given any dose $t' \geqslant t$.

Equivalently,

$$t' \geqslant t \implies y^j(t') \geqslant y^j(t).$$

Marginal potential-outcome distributions. Since for each of the T doses, there are four outcomes, the collection of marginal potential-outcome distributions under $t = 1,\dots,T$ can be represented by a $4 \times T$ column-stochastic matrix P with columns

$$P_{\bullet,t} := \begin{bmatrix} \mathbb{P}(y(t) = (0,0)) \\ \mathbb{P}(y(t) = (0,1)) \\ \mathbb{P}(y(t) = (1,0)) \\ \mathbb{P}(y(t) = (1,1)) \end{bmatrix}.$$

Parameter space. Call the space of all P matrices that are consistent with these monotonicity assumptions the parameter space, and denote it by $\mathcal{P}$.

The following lemma gives an explicit geometric characterisation of $\mathcal{P}$ as a polyhedron.

Lemma 1 (Polyhedral characterisation of $\mathcal{P}$)**.** *$P \in \mathcal{P}$ if and only if P is a column-stochastic $4 \times T$ matrix that satisfies the following inequalities, for $t = 1, \ldots, T-1$:*

$$\begin{aligned} P_{00,t} - P_{00,t+1} &\geqslant 0, \\ P_{00,t} + P_{01,t} - P_{00,t+1} - P_{01,t+1} &\geqslant 0, \\ P_{00,t} + P_{10,t} - P_{00,t+1} - P_{10,t+1} &\geqslant 0, \\ P_{11,t+1} - P_{11,t} &\geqslant 0. \end{aligned}$$

Proof. See Appendix C. □

Welfare. It is also assumed that the DM can assign expected welfare values to each of the four outcomes. These are collected in the 4-vector

$$w := \begin{bmatrix} w_{00} \\ w_{01} \\ w_{10} \\ w_{11} \end{bmatrix}.$$

The welfare values need not be additive in the two endpoints—the welfare effect of adverse effects may depend on efficacy. When welfare is nonadditive, marginal cure and adverse-effects probabilities do not generally determine expected welfare. Normalising population size to 1, it follows that the expected welfare of assigning dose t to the entire population is

$$W(t, P) = w^{\mathsf{T}} P_{\bullet,t}.$$

Denoting the standard basis vectors by $e_1, e_2, \ldots, e_T$, this can also be written as

$$W(t, P) = w^{\mathsf{T}} P e_t.$$

Similarly, the expected welfare of randomising the assigned doses according to $a \in \mathcal{A}$ is

$$W(a, P) := \sum_{t=1}^{T} a_t W(t, P) = w^{\mathsf{T}} P a.$$

Dose costs. The expected cost of assigning dose t to the entire population is denoted by $c(t)$. These costs are collected in the T-vector

$$c := (c(1), \ldots, c(T)).$$

The objective is welfare net of cost,

$$W(a, P) - c^{\mathsf{T}} a.$$

Since w and c are known ex ante and a is chosen by the DM, the unknown parameter of interest for determining optimal treatment choice is P.

Trial data. Rather than observing P directly, the DM observes the outcomes from randomised trials for K tested doses $t_1 < \cdots < t_K$, where $K \leqslant T$.[3] That is, they observe counts of individuals experiencing each outcome, for each of the tested doses. Assume that the individuals are randomly drawn from the population of interest $\mathcal{J}$. The associated data matrix is denoted by $X \in \mathbb{Z}_{\geqslant 0}^{4\times K}$ and distributed with independent columns

$$X_{\bullet,k} \sim \text{Multinom}(n_k, P_{\bullet,t_k}),$$

where $n_k \geqslant 1$ is the sample size for dose t_k. This joint experimental distribution is denoted $\mathbb{P}_{M,\mathbf{n}}$, where M is the tested-column submatrix of P defined below. If P is strictly positive in all entries of the tested columns $t_1, \ldots, t_K$, then the support of X is the set of all nonnegative integer matrices whose columns sum to $n_1, \ldots, n_K$, respectively. Denote this vector of column sums by $\mathbf{n} := (n_1, \ldots, n_K)$ and the associated support by $\mathcal{X}^{\mathbf{n}}$. It will also be useful to denote $n_{\min} := \min\{n_1, \ldots, n_K\}$. If some entries of the tested columns of P are 0, then the support of X will be a proper subset of $\mathcal{X}^{\mathbf{n}}$. Therefore, $\mathcal{X}^{\mathbf{n}}$ is exactly the set of all possible realisations of the data, for all possible values of the parameter $P \in \mathcal{P}$.

Identified subparameter. Denote by $\text{proj}_{\mathcal{M}} : \mathcal{P} \to \mathcal{M}$ the linear transformation that maps P to its column submatrix containing only the columns with indices $t_1, \ldots, t_K$. The space of such submatrices is denoted $\mathcal{M}$. It follows that the distribution of the experimental data X depends on P only through $\text{proj}_{\mathcal{M}}(P)$. Thus, while P may not be (point) identified by the sampling distribution of X, $\text{proj}_{\mathcal{M}}(P)$ is. Call $M := \text{proj}_{\mathcal{M}}(P)$ the identified subparameter of P. Then $\mathcal{M}$ is the space of all possible identified subparameters. Denote by

$$\mathcal{P}(M) := \{P \in \mathcal{P} \mid \text{proj}_{\mathcal{M}}(P) = M\}$$

the identified set for the full parameter P at the identified subparameter value M. For parsimony, I also use $\mathcal{P}$ and $\mathcal{M}$ to denote the corresponding spaces of *vectorised* parameters $\text{vec}\, P$ and $\text{vec}\, M$, where convenient.

Lemma 2 (Polyhedral characterisation of $\mathcal{M}$)**.** *$M \in \mathcal{M}$ if and only if M is a column-stochastic $4 \times K$ matrix that satisfies the following inequalities, for $k = 1, \ldots, K-1$:*

$$\begin{aligned} M_{00,k} - M_{00,k+1} &\geqslant 0, \\ M_{00,k} + M_{01,k} - M_{00,k+1} - M_{01,k+1} &\geqslant 0, \\ M_{00,k} + M_{10,k} - M_{00,k+1} - M_{10,k+1} &\geqslant 0, \\ M_{11,k+1} - M_{11,k} &\geqslant 0. \end{aligned}$$

Proof. Necessity is immediate from Lemma 1. For sufficiency, suppose that M satisfies the

[3] In practice, we will generally be interested in the incomplete-testing case $K < T$, but the results also apply to the full-testing case $K = T$.

above inequalities. Then we can construct $P \in \mathcal{P}$ satisfying $\text{proj}_{\mathcal{M}}(P) = M$ by setting

$$P_{\bullet,t} = \begin{cases} M_{\bullet,1}, & t < t_1 \\ M_{\bullet,k}, & t_k \leqslant t < t_{k+1}, \quad k = 1, \ldots, K-1 \\ M_{\bullet,K}, & t \geqslant t_K. \end{cases}$$

□

Regret Loss. The regret loss of an action $a \in \mathcal{A}$ given parameter value P is the difference between the maximum welfare net of cost attainable at P and that attained by a at P:

$$L(a, P) := \sup_{t \in \{1,\ldots,T\}} \{W(t, P) - c(t)\} - \left\{W(a, P) - c^{\mathsf{T}} a\right\}.$$

The regret loss can also be written as

$$\begin{aligned} L(a, P) &= \sup_{t \in \{1,\ldots,T\}} \left\{w^{\mathsf{T}} P e_t - c^{\mathsf{T}} e_t - w^{\mathsf{T}} P a + c^{\mathsf{T}} a\right\} \\ &= \sup_{t \in \{1,\ldots,T\}} (w^{\mathsf{T}} P - c^{\mathsf{T}})(e_t - a). \end{aligned} \tag{1}$$

Profiled regret loss. The profiled regret loss of an action $a \in \mathcal{A}$ given identified subparameter M is the maximum regret loss of action a over the set of all parameters P that are consistent with M:

$$L^{\text{prof}}(a, M) := \sup_{P \in \mathcal{P}(M)} L(a, P). \tag{2}$$

Remark 1 (Full-testing special case). If all doses were tested, $\mathcal{P}(M)$ would be a singleton $\{P\}$ and $L^{\text{prof}}(a, M)$ and $L(a, P)$ would coincide.

Population problem. The problem presented in Manski (2025) is that of choosing an action $a \in \mathcal{A}$ to minimise the profiled loss at a particular population value for the identified subparameter M.

$$v(M) := \inf_{a \in \mathcal{A}} L^{\text{prof}}(a, M).$$

Since $\mathcal{A}$ is compact and Lemma C.2 in the appendix implies that L^{prof} is continuous in a for all $M \in \mathcal{M}$, $\arg\min_{a \in \mathcal{A}} L^{\text{prof}}(a, M)$ is always nonempty.

I denote the worst-case value of the population problem over $M \in \mathcal{M}$ by

$$\overline{v} := \sup_{M \in \mathcal{M}} v(M).$$

Decision rule. A decision rule is a mapping $d : \mathcal{X}^{\mathbf{n}} \to \mathcal{A}$ that specifies a (possibly randomised) action, a, for each possible realisation of the data, X. For a given sample size vector $\mathbf{n}$, denote the set of all decision rules by $\mathcal{D}_{\mathbf{n}} := \mathcal{A}^{\mathcal{X}^{\mathbf{n}}}$. I will sometimes write $d_{\mathbf{n}}$ to emphasise that the domain of d depends on the sample size vector.[4]

[4]Since $\mathcal{X}^{\mathbf{n}}$ is finite, it is naturally equipped with the power-set σ-algebra. Then all mappings of the data into actions or estimates are trivially measurable.

Regret risk. The regret risk of a decision rule $d \in \mathcal{D}_{\mathbf{n}}$ given parameter P is the expectation over data realisations of the regret loss:

$$R_{\mathbf{n}}(d, P) := \mathbb{E}_{\text{proj}_{\mathcal{M}}(P),\mathbf{n}}\left[L(d(X), P)\right].$$

Remark 2 (Differences in terminology from Manski (2025)). In this statistical setting, there are two notions of regret: the regret *loss* is the regret associated with taking a particular action a; the regret *risk* is the expectation, over possible data realisations, of the regret loss associated with choosing a decision rule d. Since Manski (2025) focuses on the ex-post choice of a particular action and not the ex-ante problem of choosing a decision rule, he defines only regret loss and profiled regret loss. What I call "regret loss", he simply calls "regret"; and what I call "profiled regret loss", he calls "maximum regret". I avoid using the term "maximum regret" for the profiled regret loss so as to reserve the former term for the maximum regret *risk*. The term "profiled regret loss" has precedent (e.g. Xu, 2026).[5] For parsimony, I will often refer to regret loss simply as loss, profiled regret loss as profiled loss, and regret risk as risk.

Minimax regret. A rule $d^* \in \mathcal{D}_{\mathbf{n}}$ is minimax-regret (MMR) optimal if

$$\sup_{P\in\mathcal{P}} R_{\mathbf{n}}(d^*, P) = \inf_{d\in\mathcal{D}_{\mathbf{n}}} \sup_{P\in\mathcal{P}} R_{\mathbf{n}}(d, P).$$

Denote this MMR value by $V_{\mathbf{n}}$.

Expected profiled loss. The expected profiled loss of a decision rule $d \in \mathcal{D}_{\mathbf{n}}$ given identified subparameter M is

$$R^{\text{prof}}_{\mathbf{n}}(d, M) := \mathbb{E}_{M,\mathbf{n}}\left[L^{\text{prof}}(d(X), M)\right].$$

Note that if the profiled loss $L^{\text{prof}}(a, M)$ is treated as a primitive loss with respect to the identified subparameter M, then $R^{\text{prof}}_{\mathbf{n}}(d, M)$ is the corresponding risk function. In regret risk, the parameter P is held fixed across data realisations. Expected profiled loss instead allows the worst-case completion to depend on the realised data through the action $d(X)$.

Lemma 3 (Expected profiled loss bound on risk)**.** *For any decision rule $d \in \mathcal{D}_{\mathbf{n}}$ and any $M \in \mathcal{M}$,*

$$\sup_{P\in\mathcal{P}(M)} R_{\mathbf{n}}(d, P) \leqslant R^{\text{prof}}_{\mathbf{n}}(d, M).$$

Consequently,

$$\sup_{P\in\mathcal{P}} R_{\mathbf{n}}(d, P) \leqslant \sup_{M\in\mathcal{M}} R^{\text{prof}}_{\mathbf{n}}(d, M). \tag{3}$$

[5] "... the profiled regret loss function ... is defined as the maximum regret over all possible values of [the partially identified parameter] in the identified set."

Proof.

$$\sup_{P\in\mathcal{P}(M)} \mathbb{E}_{M,\mathbf{n}}\left[L(d(X),P)\right] \leqslant \mathbb{E}_{M,\mathbf{n}}\left[\sup_{P\in\mathcal{P}(M)} L(d(X),P)\right]$$
$$= \mathbb{E}_{M,\mathbf{n}}\left[L^{\text{prof}}(d(X),M)\right],$$

where the inequality follows from monotonicity of expectations. Taking the supremum over $M \in \mathcal{M}$ gives (3). □

Lemma 4 (Fixed-M problem equivalence)**.** *For any fixed $M \in \mathcal{M}$,*

$$\inf_{d\in\mathcal{D}_\mathbf{n}} R^{\text{prof}}_\mathbf{n}(d,M) = v(M) = \inf_{d\in\mathcal{D}_\mathbf{n}} \sup_{P\in\mathcal{P}(M)} R_\mathbf{n}(d,P). \tag{4}$$

Proof. See Appendix C. □

Lemma 4 implies that $v(M)$ is not just the value of the population problem at known M, but it is also the lowest attainable worst-case risk at M, and the lowest attainable expected profiled loss at M. It also implies that

$$\overline{v} \leqslant V_\mathbf{n},$$

since

$$\begin{aligned}\sup_{M\in\mathcal{M}} v(M) &= \sup_{M\in\mathcal{M}} \inf_{d\in\mathcal{D}_\mathbf{n}} \sup_{P\in\mathcal{P}(M)} R_\mathbf{n}(d,P)\\ &\leqslant \inf_{d\in\mathcal{D}_\mathbf{n}} \sup_{M\in\mathcal{M}} \sup_{P\in\mathcal{P}(M)} R_\mathbf{n}(d,P)\\ &= \inf_{d\in\mathcal{D}_\mathbf{n}} \sup_{P\in\mathcal{P}} R_\mathbf{n}(d,P).\end{aligned}$$

Remark 3 (Exact MMR rule infeasibility). In principle, an MMR optimal rule $d^*_\mathbf{n}$ can be computed for any given sample-size vector $\mathbf{n}$. This would involve optimising over the decision rules directly, which have dimension $(T-1)\cdot\#\mathcal{X}^\mathbf{n} = (T-1)\prod_{k=1}^K \binom{n_k+3}{3}$. This is computationally impractical except with the smallest of datasets. For example, if there are three doses, of which two are tested, both on samples of size 10, then the decision rules have dimension approximately equal to 1.6×10^5. Section 4 introduces a plug-in decision rule that avoids this dimensionality problem by solving a separate action-choice problem for each data realisation. Moreover, it attains MMR optimality asymptotically, and its finite-sample deviation from optimality can be bounded, given a suitable estimator of M.

4 The Plug-In Decision Rule

4.1 Definition and Properties

Plug-in decision rule. A plug-in decision rule, d^{plug}, is one which maps X to

$$d^{\text{plug}}(X) \in \arg\min_{a\in\mathcal{A}} L^{\text{prof}}(a, \widehat{M}_\mathbf{n}(X)),$$

where $\widehat{M}_{\mathbf{n}}(X)$ is some estimator of M satisfying

$$\widehat{M}_{\mathbf{n}}(X) \in \mathcal{M},$$

for all $X \in \mathcal{X}^{\mathbf{n}}$.

In the following results, $\|\cdot\|$ is an arbitrary norm on the relevant space e.g. $\mathcal{M}$. Where there are two normed spaces $\mathcal{A}$ and $\mathcal{B}$ with potentially distinct norms, the respective norms are distinguished with subscripts e.g. $\|\cdot\|_{\mathcal{A}}$. Particular ℓ^p-norms are written as $\|\cdot\|_p$.

Lemma 5. *$L^{\mathrm{prof}}(a, M)$ is uniformly Lipschitz continuous in M over a. That is, there exists $C \geqslant 0$ such that for all $a \in \mathcal{A}$ and all $M, M' \in \mathcal{M}$*

$$|L^{\mathrm{prof}}(a, M) - L^{\mathrm{prof}}(a, M')| \leqslant C\|M - M'\|.$$

The Lipschitz constant C can be chosen independently of costs c.

Proof. See Appendix C. □

The following theorem bounds how far the plug-in rule's maximum risk can exceed $V_{\mathbf{n}}$, using the worst-case expected estimation error. This makes the rule $\varepsilon_{\mathbf{n}}$-MMR (cf. Aradillas Fernández et al., 2025; Ferguson, 1967).

Theorem 1 (Plug-in regret bound)**.** *Fix T, tested doses $t_1 < \cdots < t_K$, welfare vector w, cost vector c, and a norm $\|\cdot\|$ on $\mathcal{M}$. The plug-in rule d^{plug} using estimator $\widehat{M}_{\mathbf{n}} : \mathcal{X}^{\mathbf{n}} \to \mathcal{M}$ satisfies*

$$\overline{v} \leqslant V_{\mathbf{n}} \leqslant \sup_{P \in \mathcal{P}} R_{\mathbf{n}}(d^{\mathrm{plug}}, P) \leqslant \overline{v} + \varepsilon_{\mathbf{n}} \leqslant V_{\mathbf{n}} + \varepsilon_{\mathbf{n}}, \tag{5}$$

where

$$\varepsilon_{\mathbf{n}} := 2C \cdot \sup_{M \in \mathcal{M}} \mathbb{E}_{M,\mathbf{n}} \left\| \widehat{M}_{\mathbf{n}} - M \right\|,$$

and C is the Lipschitz constant from Lemma 5 for the chosen norm.

Proof. Plug-in optimality and two applications of Lemma 5, followed by taking expectations and applying Lemma 4, give

$$R^{\mathrm{prof}}_{\mathbf{n}}(d^{\mathrm{plug}}, M) \leqslant v(M) + 2C \cdot \mathbb{E}_{M,\mathbf{n}} \|\widehat{M}_{\mathbf{n}} - M\|. \tag{6}$$

Taking the supremum over $M \in \mathcal{M}$ and applying Lemma 3 gives

$$V_{\mathbf{n}} \leqslant \sup_{P \in \mathcal{P}} R_{\mathbf{n}}(d^{\mathrm{plug}}, P) \leqslant \overline{v} + \varepsilon_{\mathbf{n}}.$$

The result then follows from $\overline{v} \leqslant V_{\mathbf{n}}$. The full proof is in Appendix C. □

Remark 4 (Comparison with Higbee (2025)). The problem studied by Higbee (2025) differs from the present one in several ways, including in restricting to nonrandomised actions. His Theorem 4.4 corresponds, in my setting, to an intermediate bound (6) in the proof of Theorem 1. The remaining steps in the proof depend on the second equality in Lemma 4 which depends in turn on convexity of the action space, i.e. availability of randomisation. Indeed, it can be shown that the analogue of (5) need not hold in his setting.

Corollary 1 (Asymptotic plug-in MMR optimality)**.** *Suppose that $\widehat{M}_{\mathbf{n}}$ is uniformly mean-consistent, in the sense that for all $\varepsilon_1 > 0$, there exists an $N_1(\varepsilon_1) \in \mathbb{N}$ such that $n_{\min} \geqslant N_1(\varepsilon_1)$ implies*

$$\sup_{M \in \mathcal{M}} \mathbb{E}_{M,\mathbf{n}} \left\| \widehat{M}_{\mathbf{n}} - M \right\| \leqslant \varepsilon_1.$$

Then for all $\varepsilon_2 > 0$, there exists an $N_2(\varepsilon_2) \in \mathbb{N}$ such that $n_{\min} \geqslant N_2(\varepsilon_2)$ implies that the plug-in rule d^{plug} is ε_2-MMR optimal, that is,

$$V_{\mathbf{n}} \leqslant \sup_{P \in \mathcal{P}} R_{\mathbf{n}}(d^{\text{plug}}, P) \leqslant V_{\mathbf{n}} + \varepsilon_2.$$

Note that Corollary 1 concerns an unscaled additive gap from minimax risk; it does not establish local asymptotic minimax optimality.

Tested-only plug-in rule. The tested-only plug-in rule d^{test} solves the plug-in problem over the restricted action space

$$\{a \in \mathcal{A} \mid a_t = 0 \text{ for all } t \notin \{t_1, \ldots, t_K\}\}.$$

Equivalently, this rule maximises estimated welfare net of cost over the tested doses, allowing mixtures among ties. In the empirical illustration of Section 5, it serves as the comparison to the unrestricted plug-in rule. It follows from the pointwise bound (6) in the proof of Theorem 1 that if

$$v(M) < v_{\text{test}}(M) := \inf \left\{ L^{\text{prof}}(a, M) \,\middle|\, a_t = 0 \text{ for all } t \notin \{t_1, \ldots, t_K\} \right\}$$

and $\varepsilon_{\mathbf{n}} < v_{\text{test}}(M) - v(M)$ then

$$\sup_{P \in \mathcal{P}(M)} R_{\mathbf{n}}(d^{\text{plug}}, P) < \sup_{P \in \mathcal{P}(M)} R_{\mathbf{n}}(d, P)$$

for any d that mixes only over tested doses, including d^{test}.

4.2 Constrained Maximum Likelihood Estimation of M

The natural first estimator of the identified subparameter M is the unconstrained maximum likelihood estimator, $\widehat{M}^u_{\mathbf{n}}$, which is just the matrix of empirical outcome frequencies at each tested dose. However, these empirical frequencies may violate the monotonicity assumptions

of Lemma 2, even if the true M does not. Such a violation makes the plug-in problem infeasible.

Therefore, in order to ensure feasibility of the plug-in problem, I propose the use of a constrained maximum likelihood estimator, which maximises the likelihood over the constraint set $\mathcal{M}$. $\mathcal{M}$ is defined by the $4(K-1)$ inequalities in Lemma 2, and the requirement that each column of M lie in the 3-simplex.

Constrained MLE of M. A constrained MLE of M is any selection $\widehat{M}^c_{\mathbf{n}} : \mathcal{X}^{\mathbf{n}} \to \mathcal{M}$ satisfying

$$\widehat{M}^c_{\mathbf{n}}(X) \in \arg\max_{m \in \mathcal{M}} \sum_{k=1}^{K} \sum_{y \in \mathcal{Y}} X_{y,k} \log m_{y,k}, \tag{7}$$

with the convention

$$x \log z := \begin{cases} 0, & x = 0, z \geqslant 0, \\ x \log z, & x > 0, z > 0, \\ -\infty, & x > 0, z = 0. \end{cases}$$

Since $\mathcal{M}$ is compact and the objective is upper semicontinuous, the argmax correspondence is nonempty-valued.

Let $\|\cdot\|_1$ denote the entrywise ℓ^1 norm on matrices, i.e.

$$\|M\|_1 := \|\operatorname{vec} M\|_1 = \sum_{k=1}^{K} \sum_{y \in \mathcal{Y}} |M_{y,k}|.$$

Proposition 1 (Uniform bound for mean error of constrained MLE)**.**

$$\sup_{M \in \mathcal{M}} \mathbb{E}_{M,\mathbf{n}} \left\| \widehat{M}^c_{\mathbf{n}} - M \right\|_1 \leqslant K \frac{\sqrt{6} + \sqrt{3}}{\sqrt{n_{\min}}}.$$

Proof. See Appendix C. □

Combining Theorem 1 with Proposition 1 gives the following bound for the plug-in rule using the constrained MLE $\widehat{M}^c_{\mathbf{n}}$ and the ℓ^1 norm on $\mathcal{M}$:

$$\varepsilon^c_{\mathbf{n}} \leqslant 2C_1 K \frac{\sqrt{6} + \sqrt{3}}{\sqrt{n_{\min}}},$$

where C_1 is the Lipschitz constant from Lemma 5 for the particular case of an ℓ^1-norm, and K is the number of tested doses. Hence

$$\varepsilon^c_{\mathbf{n}} = O\left(n_{\min}^{-1/2}\right).$$

4.3 Linear Program Representation and Plug-In Computation

The population problem at M, and hence the plug-in problem using any estimate $\widehat{M} \in \mathcal{M}$, can be written as a single linear program and solved using standard methods. This uses the threshold representation in Appendix B: S maps the threshold distribution into outcome probabilities, and Ψ selects the outcome probabilities at the tested doses.

Proposition 2 (LP representation of plug-in problem)**.** *Let* $\mathbf{m} := \operatorname{vec} M$*. Then*

$$\inf_{a \in \mathcal{A}} L^{\text{prof}}(a, \mathbf{m}) \tag{8}$$

can equivalently be written as the linear program

$$\begin{aligned} \inf_{a, z, (u_t, \lambda_t)_{t=1}^T} \quad & z \\ \text{s.t.} \quad & z - \mathbf{m}^\mathsf{T} u_t - \lambda_t \geqslant -c^\mathsf{T}(e_t - a) && t = 1, \ldots, T \\ & S^\mathsf{T} \Psi^\mathsf{T} u_t + \lambda_t \mathbf{1} + Ra \geqslant r_t && t = 1, \ldots, T \\ & \mathbf{1}^\mathsf{T} a = 1 \\ & a \geqslant 0, \end{aligned} \tag{9}$$

where $u_t \in \mathbb{R}^{4K}$ *and* $\lambda_t \in \mathbb{R}$ *are free variables for* $t = 1, \ldots, T$*;* S *and* Ψ *are binary matrices defined in Appendix B;* $r_t := S^\mathsf{T} \operatorname{vec}(we_t^\mathsf{T}) \in \mathbb{R}^{(T+1)^2}$ *for* $t = 1, \ldots, T$*; and* $R \in \mathbb{R}^{(T+1)^2 \times T}$ *is the matrix with columns given by* r_1*,* r_2*,* $\ldots$*,* r_T*.*

Proof. Lemma B.1 in the appendix shows that for fixed a the profiled loss can be written as the envelope over T equality-constrained maximisation linear programs. Dualising the inner linear program for fixed t transforms it into an equivalent inequality-constrained minimisation problem. The inequality-constrained form allows the introduction of an epigraph variable z so that the profiled loss itself can be written as a single linear program. The outer problem of minimising the value of this linear program with respect to a is then also a linear program. Details are given in Appendix C. □

The following algorithm summarises the steps required to compute the plug-in:

Algorithm 1 Plug-in computation

Require: Number of doses T; tested doses $\{t_1, \ldots, t_K\} \subseteq \{1, \ldots, T\}$; data X; welfare vector w; cost vector c.

1: Compute the constrained maximum likelihood estimator $\widehat{M}^c(X)$ as in Section 4.2.

2: Compute the matrix S from T, and the matrix Ψ from T and $\{t_1, \ldots, t_K\}$, as in Appendix B.

3: Compute $r_t := S^\mathsf{T} \operatorname{vec}(we_t^\mathsf{T})$, for $t = 1, \ldots, T$; and $R := \begin{bmatrix} r_1 & \cdots & r_T \end{bmatrix}$.

4: Construct and solve the linear program (9) with $\mathbf{m} := \operatorname{vec} \widehat{M}^c(X)$ for optimal action $d^{\text{plug}}(X)$.

5 Empirical Illustration

I illustrate the plug-in rule using data from two studies on oseltamivir, a medication used to treat and prevent influenza. In both studies, eligible participants were non-immunised and otherwise healthy adults aged 18–65 with febrile influenza-like illness of no more than 36 hours' duration. Participants were randomly assigned to a twice-daily five-day course of oseltamivir at one of three doses: 0 mg (placebo), 75 mg, or 150 mg. The study reported by Treanor et al. (2000) was conducted at centres across the United States, while that reported by Nicholson et al. (2000) was conducted at centres in Europe, Canada, and Hong Kong. The primary efficacy endpoint was time from treatment initiation to resolution of illness, as determined by symptom alleviation. To apply my method, I discretise this endpoint into four binned resolution times (in hours): $\leqslant 48$, $(48, 96]$, $(96, 168]$, and > 168. I use a binary adverse-effects endpoint: whether a participant experienced nausea or vomiting. Since there are more than two levels for the efficacy endpoint, I apply the generalisation of my model to endpoints with more than two levels (see Appendix D). I use participants irrespective of laboratory confirmation of influenza. After excluding 13 participants listed in the clinical study reports as having received treatment without randomisation, and omitting 156 participants for whom the joint outcome was incompletely observed, the pooled data consist of 1,186 individuals: 388 assigned the placebo, 396 assigned the 75 mg dose, and 402 assigned the 150 mg dose. The dose menu available to clinicians is larger: 30 mg and 45 mg capsules are also available and a dose can be administered using more than one capsule at a time (Roche Products Limited, 2021). I thus assume that the full dose menu consists of 0 mg and all multiples of 15 mg between 30 and 150 mg inclusive.

The application is intended to illustrate my paper's method, and should not be interpreted as making policy recommendations or as contributing to the wider debate over the benefits and harms of oseltamivir (Jefferson et al., 2014b). In particular, the model imposes monotone dose-response by assumption, and the results are highly sensitive to the choice of welfare values w and the medication cost. In the baseline illustration I calibrate the benefits of earlier illness resolution using QALY valuations from the National Institute for Health and Care Excellence (NICE, 2026, 2009), which imply a welfare gain of approximately £18.49 per day of earlier illness resolution. I assume nausea or vomiting welfare penalties ranging from £20.14 to £27.74, greater when they coincide with longer influenza-symptom persistence. Medication costs are based on the cheapest capsule combination for each dose, using inflation-adjusted NHS list prices (UKHSA, 2026, Table 9). Appendix E provides the details of these calculations. The implied welfare values for different outcomes are listed in Table 2.

Time to resolution of illness (hours)	Adverse effects	Welfare (£)	Estimated probability		
			Placebo	75 mg	150 mg
> 168	No	0.00	0.216	0.134	0.134
	Yes	-27.74	0.026	0.070	0.021
$(96, 168]$	No	76.42	0.253	0.181	0.167
	Yes	52.50	0.018	0.044	0.065
$(48, 96]$	No	116.96	0.265	0.233	0.246
	Yes	95.07	0.023	0.048	0.070
$\leqslant 48$	No	152.08	0.191	0.242	0.242
	Yes	131.94	0.008	0.048	0.055

Table 2: Baseline welfare values and constrained maximum likelihood estimates of joint outcome probabilities, using 1,186 retained participants. Welfare is in pounds per person, before medication costs, normalised to zero for the outcome in which resolution time is longer than 168 hours and no adverse effects are experienced. Adverse effects denote nausea or vomiting.

I apply the plug-in rule based on constrained maximum likelihood estimates of the outcome probabilities, as described in Sections 4.2 and 4.3 and Appendix D. Recall that these impose monotone dose-response assumptions on both efficacy and adverse-effects incidence. The estimated outcome probabilities are given in Table 2. The monotonicity constraints between the placebo and 75 mg doses are satisfied by the empirical frequencies; several of the corresponding constraints linking the 75 and 150 mg doses bind.

The plug-in rule randomises between the placebo (0.86) and the 30 mg dose (0.14), attaining estimated profiled regret loss £1.60. Restricting the allocation to tested doses, the plug-in rule places all probability mass on the placebo, attaining estimated profiled regret loss £1.86 against the full menu. Therefore, the unrestricted plug-in rule reduces estimated profiled regret loss by £0.26 per person, or 13.9%.

With the estimated tested-dose probabilities held fixed, welfare net of cost is £82.81 for placebo and can range from £71.30 to £84.67 for the 30 mg dose. Relative to placebo, the 30 mg dose can therefore give a gain of about £1.86 or a loss of about £11.51. The mixture balances the regret from missing the possible gain against the regret from incurring the possible loss.

Figure 1 shows how the plug-in dose mixture changes as medication cost (left) and the welfare decrement of adverse effects (right) vary. Multiplying medication cost by any given positive factor is equivalent to dividing the monetary value of a QALY by that same factor.

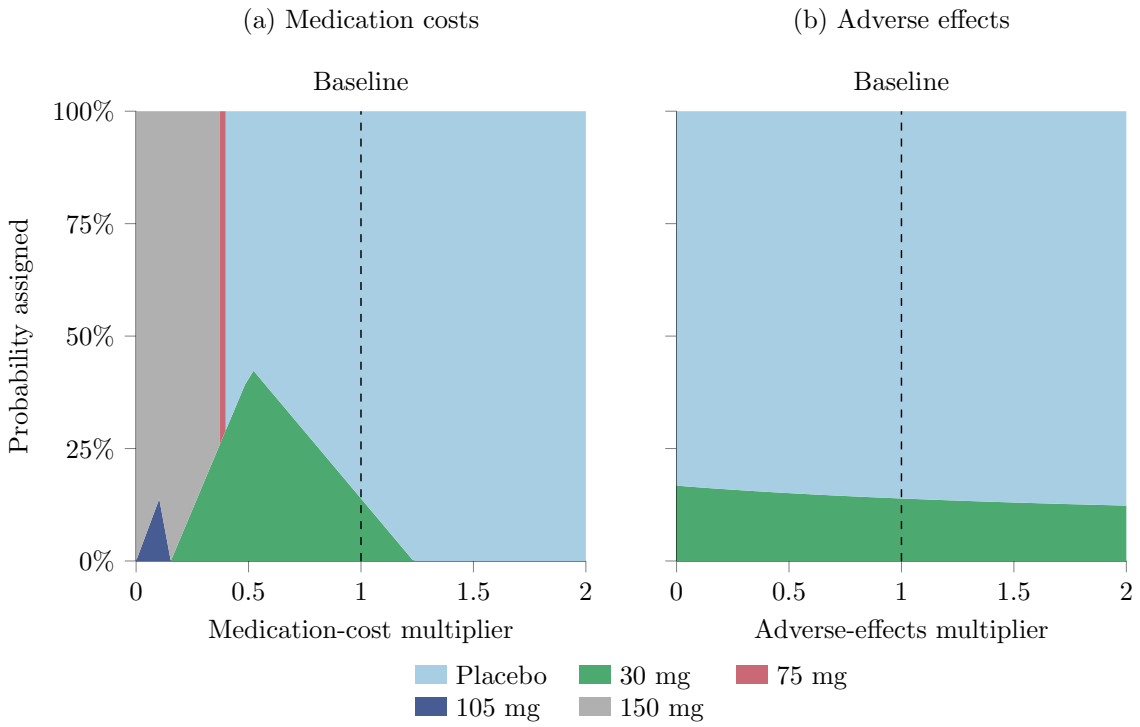


Figure 1: Plug-in dose allocations at different medication costs and adverse-effect welfare decrements. A multiplier of one denotes the baseline. All other inputs are held fixed.

6 Conclusion

In this paper, I extended the dose-choice problem of Manski (2025) to a finite-sample statistical setting. I studied a plug-in rule that treats estimated outcome probabilities at the tested doses as if they were the true values and solves the resulting population problem. For a uniformly mean-consistent estimator of the identified subparameter M, this plug-in rule asymptotically minimises worst-case risk. In particular, when a constrained maximum likelihood estimator is used, the difference between the maximum risk of the plug-in rule and the minimax risk vanishes at a rate no slower than $n_{\min}^{-1/2}$, where $n_{\min}$ is the smallest per-arm sample size. Moreover, given an estimate of the subparameter, the plug-in problem can be reformulated as a single linear program, allowing it to be computed efficiently. I illustrated the plug-in rule using data from two studies on oseltamivir. Under the baseline calibration, it attains lower estimated profiled regret loss than the best mixture over tested doses.

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A Notation

Notation	Meaning
t, T	dose index and number of doses
$a, \mathcal{A}$	mixture over doses and simplex of dose mixtures
t_k, K	k-th tested dose and number of tested doses
$y, \mathcal{Y}$	outcome and outcome set
$P, \mathcal{P}$	outcome probability matrix and parameter space
$P_{\bullet,t}, P_{y,\bullet}$	column for dose t and row for outcome y
w, W	welfare vector and welfare function
$c(t), c$	cost of dose t, and vector of dose costs
$X, \mathcal{X}^{\mathbf{n}}$	data matrix and space of possible data realisations
$n_k, \mathbf{n}, n_{\min}$	sample size of arm k, vector of arm sizes, and smallest arm size
$M, \mathcal{M}$	tested-dose submatrix of P and identified subparameter space
$\text{proj}_{\mathcal{M}}$	projection from $\mathcal{P}$ onto $\mathcal{M}$
$\mathcal{P}(M)$	identified set for P at M
$\mathbb{P}_{M,\mathbf{n}}, \mathbb{E}_{M,\mathbf{n}}$	sampling distribution of X and expectation under that distribution
$d_{\mathbf{n}}, \mathcal{D}_{\mathbf{n}}$	decision rule and space of possible decision rules
L, L^{prof}	regret loss and profiled regret loss
$R_{\mathbf{n}}, R^{\text{prof}}_{\mathbf{n}}$	regret risk and expected profiled regret loss
$V_{\mathbf{n}}$	minimax regret risk
$v(M)$	minimum profiled loss at M
$\overline{v}$	$\sup_{M \in \mathcal{M}} v(M)$
$d^{\text{plug}}_{\mathbf{n}}$	plug-in rule
$\widehat{M}^u_{\mathbf{n}}, \widehat{M}^c_{\mathbf{n}}$	unconstrained and constrained MLEs of M

Table 3: Key notation.

B Threshold-Distribution Form of Profiled Regret Loss

Assumption 1 (monotone dose-response functions) implies that, for any particular individual j, the two dose-response functions can be written as step functions, each defined by a single threshold dose in $\{1, \ldots, T+1\}$:

$$y^j_e(t) = \mathbf{1}\{t \geqslant \theta_{e,j}\},$$
$$y^j_{\text{ae}}(t) = \mathbf{1}\{t \geqslant \theta_{\text{ae},j}\}.$$

If a threshold is 1, then the associated outcome is 1 for all doses; if a threshold is $T+1$, then the associated outcome is 0 for all doses.

Threshold distribution. Each individual's pair of dose-response functions is determined by their two threshold doses. The population distribution of these functions can therefore be represented by the joint probability mass function

$$Q : \{1, \dots, T+1\}^2 \to [0, 1].$$

Also write Q for the matrix with entries $Q_{ij} := Q(i, j)$. Note that each threshold distribution Q specifies a unique collection of marginal potential-outcome distributions P,[6] by the equalities:

$$\begin{aligned}
\mathbb{P}(y(t) = (0, 0)) &= \mathbb{P}(\theta_e > t, \theta_{\text{ae}} > t), \\
\mathbb{P}(y(t) = (0, 1)) &= \mathbb{P}(\theta_e > t, \theta_{\text{ae}} \leqslant t), \\
\mathbb{P}(y(t) = (1, 0)) &= \mathbb{P}(\theta_e \leqslant t, \theta_{\text{ae}} > t), \\
\mathbb{P}(y(t) = (1, 1)) &= \mathbb{P}(\theta_e \leqslant t, \theta_{\text{ae}} \leqslant t).
\end{aligned} \tag{10}$$

The equalities in (10) imply that the mapping $Q \mapsto P$ is linear.

For $t \in \{1, \dots, T\}$ and $\theta = (\theta_e, \theta_{\text{ae}}) \in \{1, \dots, T+1\}^2$, denote

$$y(t, \theta) := \big(\mathbf{1}\{t \geqslant \theta_e\}, \mathbf{1}\{t \geqslant \theta_{\text{ae}}\}\big).$$

Order outcomes y and threshold pairs θ lexicographically.[7] Use these orders for the rows and columns, respectively, of $s(t) \in \{0, 1\}^{4 \times (T+1)^2}$, which is defined by

$$s(t)_{y,\theta} := \mathbf{1}\{y(t, \theta) = y\}, \qquad y \in \mathcal{Y}, \quad \theta \in \{1, \dots, T+1\}^2.$$

Then the equalities in (10) can be written as

$$P_{\bullet,t} = s(t) \operatorname{vec}(Q^{\mathsf{T}}).$$

It follows that

$$\mathbf{p} = S\mathbf{q},$$

where $\mathbf{p} = \operatorname{vec} P$, $\mathbf{q} := \operatorname{vec}(Q^{\mathsf{T}})$, and S is the $4T \times (T+1)^2$ matrix

$$S := \begin{bmatrix} s(1) \\ s(2) \\ \vdots \\ s(T) \end{bmatrix}.$$

[6] The converse is not true: distinct threshold distributions can specify the same marginal potential-outcome distributions. For example if $T = 2$, the uniform distributions over $\{(1, 3), (2, 2)\}$ and $\{(1, 2), (2, 3)\}$ both induce $P_{\bullet,1} = (1/2, 0, 1/2, 0)$ and $P_{\bullet,2} = (0, 0, 1/2, 1/2)$. This is because Q contains sufficient information to specify the joint distribution of potential outcomes, whereas P contains only the marginal potential-outcome distributions.

[7] For y this means $(0, 0), (0, 1), (1, 0), (1, 1)$; for θ it means $(1, 1), (1, 2), \dots, (1, T+1), \dots, (T+1, 1), \dots, (T+1, T+1)$.

In Section 3, $\text{proj}_{\mathcal{M}} : \mathcal{P} \to \mathcal{M}$ denoted the mapping that selects columns $t_1, \ldots, t_K$ from P. This can be represented as post-multiplication by a $T \times K$ binary matrix C that has ones only in entries (t_k, k) for $k = 1, \ldots, K$. Then $M = PC$. It follows that $\mathbf{m} := \text{vec}\, M = \text{vec}(PC) = \text{vec}(IPC) = (C^\mathsf{T} \otimes I)\mathbf{p}$ where $\otimes$ denotes the Kronecker product and the last equality is Theorem A.7.9 in Hansen (2022). Then denote $\Psi := C^\mathsf{T} \otimes I$ so that $\mathbf{m} = \Psi\mathbf{p}$.

The following lemma characterises the profiled loss in terms of the threshold distribution Q.

Lemma B.1.

$$\begin{aligned} L^{\text{prof}}(a, \mathbf{m}) = \sup_{t \in \{1,\ldots,T\}} \sup_{\mathbf{q}} \quad & \text{vec}(w(e_t - a)^\mathsf{T})^\mathsf{T} S\mathbf{q} - c^\mathsf{T}(e_t - a) \\ s.t.\ \ & \Psi S\mathbf{q} = \mathbf{m} \\ & \mathbf{1}^\mathsf{T}\mathbf{q} = 1 \\ & \textit{and}\ \mathbf{q} \geqslant 0, \end{aligned}$$

where $\mathbf{1}$ *is the sum vector, and* S *and* Ψ *are the binary matrices defined above.*

Proof. From equations (1) and (2), the profiled loss is

$$\sup_{P \in \mathcal{P}(M)} \sup_{t \in \{1,\ldots,T\}} w^\mathsf{T} P(e_t - a) - c^\mathsf{T}(e_t - a).$$

Write[8]

$$w^\mathsf{T} P(e_t - a) = \text{vec}(w(e_t - a)^\mathsf{T})^\mathsf{T}\mathbf{p}.$$

By the discussion above, $\mathbf{p} \in \mathcal{P}(M)$ if and only if

$$\Psi\mathbf{p} = \mathbf{m}$$

and

$$\mathbf{p} = S\mathbf{q}$$

for some $\mathbf{q} \in \Delta^{(T+1)^2-1}$. □

C Proofs

Proof of Lemma 1. Column stochasticity is equivalent to each column being a valid probability distribution.

Consider the set $\{0,1\}^2$ equipped with the canonical partial order $(a_1, b_1) \leqslant (a_2, b_2) \iff a_1 \leqslant a_2$ and $b_1 \leqslant b_2$. I claim that the $4(T-1)$ inequalities in Lemma 1 are exactly

[8] By parts 4 and 9 of Theorem A.7 in Hansen (2022), $x^\mathsf{T} A y = \text{vec}(x^\mathsf{T} A y) = (y^\mathsf{T} \otimes x^\mathsf{T})\,\text{vec}\, A = (y \otimes x)^\mathsf{T}\,\text{vec}\, A = \text{vec}(xy^\mathsf{T})^\mathsf{T}\,\text{vec}\, A$.

those required for the columns of P to be ordered by stochastic dominance. The nontrivial lower sets of $\{0,1\}^2$ are $\{(0,0)\}$, $\{(0,0),(0,1)\}$, $\{(0,0),(1,0)\}$, and $\{(0,0),(0,1),(1,0)\}$. Therefore, $P_{\bullet,t+1}$ stochastically dominates $P_{\bullet,t}$ if and only if $P_{\bullet,t}$ assigns a weakly greater probability to each of these lower sets. It is clear that this is true for all t if and only if the four inequalities in Lemma 1 hold for t.

It is also clear that if all dose-response functions in the population are monotone, then the marginal potential-outcome distributions must be ordered by stochastic dominance in this way. For example, if a fraction λ of the population $\mathcal{J}$ does not experience adverse effects at dose t, then the fraction that does not experience adverse effects at dose $t+1$ can be no more than λ.

Therefore, it remains only to prove the converse: if a column-stochastic $P \in \mathbb{R}^{4\times T}$ satisfies the $4(T-1)$ inequalities in the lemma, then P is consistent with Assumption 1. Following the discussion in Appendix B, it suffices to show that any such P can be generated by some threshold distribution Q.

Strassen's theorem (Klenke, 2020, Theorem 17.59) then implies that there exists a coupling of $y(t-1)$ and $y(t)$, Π_t, with marginal distributions given by $P_{\bullet,t-1}$ and $P_{\bullet,t}$ respectively, such that

$$y(t-1) \leqslant y(t) \quad \Pi_t\text{-a.s.} \tag{11}$$

Conditionalise each of these joint distributions to get kernels $K_t(y' \mid y) := \Pi_t(y,y')/P_{y,t-1}$ for $y, y' \in \mathcal{Y}$, $t = 2,\dots,T$, and $P_{y,t-1} > 0$, assigning an arbitrary probability distribution to $K_t(\cdot \mid y)$ when $P_{y,t-1} = 0$. Then, by letting $y(1) \sim P_{\bullet,1}$ and iteratively generating $y(t) \sim K_t(\cdot \mid y(t-1))$, we obtain a full joint distribution over all dose-outcomes, with marginals given by the columns of P. Moreover, this joint distribution has the ordering property (11) for all $t \in \{2,\dots,T\}$. Clearly this joint distribution over $\{y(t) \mid t = 1,\dots,T\}$ is equivalent to a distribution over dose-response functions $y(\cdot)$. Moreover, these dose-response functions are almost surely monotone, so we are done. □

Proof of Lemma 4. Consider the first equality in (4).

$$\begin{aligned}
\inf_{d\in\mathcal{D}_{\mathbf{n}}} R^{\text{prof}}_{\mathbf{n}}(d,M) &= \inf_{d\in\mathcal{D}_{\mathbf{n}}} \mathbb{E}_{M,\mathbf{n}}\left[\sup_{P\in\mathcal{P}(M)} L(d(X),P)\right] \\
&\geqslant \inf_{a\in\mathcal{A}} \mathbb{E}_{M,\mathbf{n}}\left[\sup_{P\in\mathcal{P}(M)} L(a,P)\right] \\
&= \inf_{a\in\mathcal{A}} \sup_{P\in\mathcal{P}(M)} L(a,P).
\end{aligned}$$

This inequality binds if d is the constant rule that chooses $a^* \in \arg\min_{a\in\mathcal{A}} L^{\text{prof}}(a,M)$.

Now consider the second equality in (4). Define the average action of decision rule d under M,

$$\overline{a}_d(M) := \mathbb{E}_{M,\mathbf{n}}\left[d(X)\right].$$

This is itself a valid action, by convexity of $\mathcal{A}$.

Observe that $L(a, P)$ is affine in a at each fixed P since

$$L(a, P) = -(w^\mathsf{T} P - c^\mathsf{T})a + \sup_t (w^\mathsf{T} P - c^\mathsf{T}) e_t.$$

Thus

$$\begin{aligned} R_{\mathbf{n}}(d, P) &= \mathbb{E}_{M,\mathbf{n}}[L(d(X), P)] \\ &= L(\mathbb{E}_{M,\mathbf{n}}[d(X)], P) \\ &= L(\overline{a}_d(M), P). \end{aligned}$$

Taking the supremum over $P \in \mathcal{P}(M)$,

$$\sup_{P \in \mathcal{P}(M)} R_{\mathbf{n}}(d, P) = L^{\text{prof}}(\overline{a}_d(M), M) \geqslant \inf_a L^{\text{prof}}(a, M). \tag{12}$$

But any constant rule d_a that always chooses action a is a valid decision rule and

$$\begin{aligned} \sup_{P \in \mathcal{P}(M)} R_{\mathbf{n}}(d_a, P) &= \sup_{P \in \mathcal{P}(M)} \mathbb{E}_{M,\mathbf{n}}[L(d_a(X), P)] \\ &= \sup_{P \in \mathcal{P}(M)} \mathbb{E}_{M,\mathbf{n}}[L(a, P)] \\ &= \sup_{P \in \mathcal{P}(M)} L(a, P) \\ &= L^{\text{prof}}(a, M), \end{aligned}$$

hence

$$\inf_d \sup_{P \in \mathcal{P}(M)} R_{\mathbf{n}}(d, P) \leqslant \inf_a L^{\text{prof}}(a, M).$$

Combining with (12),

$$\inf_d \sup_{P \in \mathcal{P}(M)} R_{\mathbf{n}}(d, P) = \inf_a L^{\text{prof}}(a, M).$$

□

The following results use the concept of a dual norm $\|\cdot\|_*$, which is defined with respect to a given norm $\|\cdot\|$ by

$$\|z\|_* := \sup\{z^\mathsf{T} x \mid \|x\| \leqslant 1\}.$$

This definition immediately implies the dual norm inequality

$$|z^\mathsf{T} x| \leqslant \|x\| \, \|z\|_*.$$

Lemma C.1. *Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a convex and compact polyhedron, $A \in \mathbb{R}^{m \times n}$, and $\mathcal{C} \subseteq \mathbb{R}^n$ be compact. For $b \in \mathbb{R}^m$, define*

$$\Gamma(b) := \{x \in \mathcal{X} \mid Ax = b\},$$

and let $\mathcal{B}$ be the effective domain of Γ,

$$\mathcal{B} := \{b \in \mathbb{R}^m \mid \Gamma(b) \neq \varnothing\}.$$

Define

$$v(b, c) := \sup_{x \in \Gamma(b)} c^{\mathsf{T}} x.$$

Then there exists a constant $L \in [0, \infty)$ such that

$$|v(b, c) - v(b', c)| \leqslant L\|c\|_* \, \|b - b'\|_{\mathcal{B}},$$

for all $b, b' \in \mathcal{B}$ and all $c \in \mathcal{C}$. Consequently, if $\sup_{c \in \mathcal{C}} \|c\|_* < \infty$, then $v(\cdot, c)$ is uniformly Lipschitz in b over $c \in \mathcal{C}$.

Proof. First note that

$$\operatorname{Gr} \Gamma = \{(b, x) \mid b \in \mathbb{R}^m, x \in \mathcal{X}, Ax = b\}$$

is a convex polyhedron. It follows that in this case Γ is a Lipschitz continuous correspondence in the sense that there exists $L \geqslant 0$ such that

$$d_H(\Gamma(b), \Gamma(b')) \leqslant L\|b - b'\|_{\mathcal{B}},$$

for all $b, b' \in \mathcal{B}$, where d_H is the Hausdorff distance induced by $\|\cdot\|_{\mathcal{X}}$ (Dontchev and Rockafellar, 2009, Theorem 3C.3)[9]. Assume, without loss of generality, that $v(b, c) \geqslant v(b', c)$. Choose $x^* \in \arg\max_{x \in \Gamma(b)} c^{\mathsf{T}} x$ and $x' \in \Gamma(b')$ satisfying $\|x^* - x'\|_{\mathcal{X}} = d(x^*, \Gamma(b')) := \inf_{x \in \Gamma(b')} \|x^* - x\|_{\mathcal{X}}$. Then

$$\begin{aligned}
|v(b, c) - v(b', c)| &= c^{\mathsf{T}} x^* - v(b', c) \\
&\leqslant c^{\mathsf{T}}(x^* - x') \\
&\leqslant \|c\|_* \|x^* - x'\|_{\mathcal{X}} \\
&= \|c\|_* \, d(x^*, \Gamma(b')) \\
&\leqslant \|c\|_* \, d_H(\Gamma(b), \Gamma(b')) \\
&\leqslant L\|c\|_* \, \|b - b'\|_{\mathcal{B}},
\end{aligned}$$

where $\|\cdot\|_*$ is the norm dual to $\|\cdot\|_{\mathcal{X}}$. □

The set $\mathcal{M}$ is the image of the compact simplex $\Delta^{(T+1)^2-1}$ under the linear transformation $\mathbf{q} \mapsto \mathbf{m}$ defined by $\mathbf{m} = \Psi S \mathbf{q}$, and is therefore itself compact.

[9] Theorem 3C.3 uses the Euclidean norm, but the result holds for a general norm by finite-dimensional norm equivalence.

Proof of Lemma 5. For any $t \in \{1, \ldots, T\}$, define the inner value function:

$$\begin{aligned} v(a, t, \mathbf{m}) := \sup_{\mathbf{q}} \ & \operatorname{vec}(w(e_t - a)^\mathsf{T})^\mathsf{T} S\mathbf{q} \\ \text{s.t. } & \Psi S\mathbf{q} = \mathbf{m} \\ & \mathbf{1}^\mathsf{T}\mathbf{q} = 1 \\ & \text{and } \mathbf{q} \geqslant 0. \end{aligned}$$

Lemma C.1 implies that this value function is Lipschitz in $\mathbf{m}$ uniformly over a. To apply the lemma, note that the feasible sets

$$\Gamma(\mathbf{m}) := \{\mathbf{q} \in \Delta^{(T+1)^2-1} \mid \Psi S\mathbf{q} = \mathbf{m}\}$$

are nonempty for all $\mathbf{m} \in \mathcal{M}$ and contained in the simplex, and that the set of coefficients $\{\operatorname{vec}(w(e_t - a)^\mathsf{T})^\mathsf{T} S \mid a \in \mathcal{A}\}$ is compact. Write

$$|v(a, t, \mathbf{m}) - v(a, t, \mathbf{m}')| \leqslant C(t)\|\mathbf{m} - \mathbf{m}'\|,$$

for some $C(t) \geqslant 0$, and all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}$, $a \in \mathcal{A}$. By Lemma B.1,

$$L^{\text{prof}}(a, \mathbf{m}) = \sup_{t=1,\ldots,T} \{v(a, t, \mathbf{m}) - c^\mathsf{T}(e_t - a)\}.$$

Choosing $C := \max_{t=1,\ldots,T} C(t)$,

$$\begin{aligned} |L^{\text{prof}}(a, \mathbf{m}) - L^{\text{prof}}(a, \mathbf{m}')| &\leqslant \sup_{t=1,\ldots,T} |v(a, t, \mathbf{m}) - v(a, t, \mathbf{m}')| \\ &\leqslant C\|\mathbf{m} - \mathbf{m}'\|, \end{aligned}$$

for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}$ and all $a \in \mathcal{A}$, as required. □

We also have continuity of the profiled loss in a.

Lemma C.2. *$L^{\text{prof}}(a, \mathbf{m})$ is continuous in a at all $\mathbf{m} \in \mathcal{M}$.*

Proof. Fix $\mathbf{m} \in \mathcal{M}$ and $a, a' \in \mathcal{A}$. For any $P \in \mathcal{P}(\mathbf{m})$, the maximum welfare net of cost attainable at P is the same for both actions, so

$$\begin{aligned} |L(a, P) - L(a', P)| &= |(w^\mathsf{T} P - c^\mathsf{T})(a' - a)| \\ &\leqslant \left(\|P^\mathsf{T} w\|_\infty + \|c\|_\infty\right) \|a' - a\|_1 \\ &\leqslant (\|w\|_\infty + \|c\|_\infty)\|a' - a\|_1, \end{aligned}$$

where the first inequality applies the triangle and dual norm inequalities; and the second inequality uses column stochasticity of P. Thus,

$$\begin{aligned} |L^{\text{prof}}(a, \mathbf{m}) - L^{\text{prof}}(a', \mathbf{m})| &\leqslant \sup_{P \in \mathcal{P}(\mathbf{m})} |L(a, P) - L(a', P)| \\ &\leqslant (\|w\|_\infty + \|c\|_\infty)\|a' - a\|_1. \end{aligned}$$

□

Proof of Theorem 1. Fix $\mathbf{n} \in \mathbb{N}^K$. Let $d \in \mathcal{D}_{\mathbf{n}}$ be an arbitrary decision rule. Using uniform Lipschitz continuity of $L^{\text{prof}}(a, M)$ in M over a, and the definition of d^{plug} as a plug-in rule,

$$\begin{aligned} L^{\text{prof}}(d^{\text{plug}}(X), M) &\leqslant L^{\text{prof}}(d^{\text{plug}}(X), \widehat{M}_{\mathbf{n}}) + C\|\widehat{M}_{\mathbf{n}} - M\| \\ &\leqslant L^{\text{prof}}(d(X), \widehat{M}_{\mathbf{n}}) + C\|\widehat{M}_{\mathbf{n}} - M\| \\ &\leqslant L^{\text{prof}}(d(X), M) + 2C\|\widehat{M}_{\mathbf{n}} - M\|. \end{aligned}$$

Taking expectations and the infimum over $d \in \mathcal{D}_{\mathbf{n}}$,

$$R^{\text{prof}}_{\mathbf{n}}(d^{\text{plug}}, M) \leqslant \inf_{d \in \mathcal{D}_{\mathbf{n}}} R^{\text{prof}}_{\mathbf{n}}(d, M) + 2C \cdot \mathbb{E}_{M,\mathbf{n}}\|\widehat{M}_{\mathbf{n}} - M\|.$$

The first equality in Lemma 4 implies

$$R^{\text{prof}}_{\mathbf{n}}(d^{\text{plug}}, M) \leqslant v(M) + 2C \cdot \mathbb{E}_{M,\mathbf{n}}\|\widehat{M}_{\mathbf{n}} - M\|,$$

and so

$$\sup_{M \in \mathcal{M}} R^{\text{prof}}_{\mathbf{n}}(d^{\text{plug}}, M) \leqslant \overline{v} + 2C \cdot \sup_{M \in \mathcal{M}} \mathbb{E}_{M,\mathbf{n}}\|\widehat{M}_{\mathbf{n}} - M\|.$$

By Lemma 3,

$$V_{\mathbf{n}} := \inf_{d \in \mathcal{D}_{\mathbf{n}}} \sup_{P \in \mathcal{P}} R_{\mathbf{n}}(d, P) \leqslant \sup_{P \in \mathcal{P}} R_{\mathbf{n}}(d^{\text{plug}}, P) \leqslant \sup_{M \in \mathcal{M}} R^{\text{prof}}_{\mathbf{n}}(d^{\text{plug}}, M).$$

The result now follows from $\overline{v} \leqslant V_{\mathbf{n}}$. □

For any two discrete distributions p and q, define the Kullback-Leibler (KL) divergence between p and q by

$$D_{\text{KL}}(p\|q) := \begin{cases} +\infty, & q_i = 0 < p_i \text{ for some } i \\ \sum_{\text{supp}\, q} p_i \log\left(\frac{p_i}{q_i}\right), & \text{otherwise,} \end{cases}$$

with the convention that $0 \log(0/q_i) = 0$ for all $q_i \geqslant 0$. The following result says that the constrained MLE is the arm-size-weighted reverse-KL projection of the MLE (empirical distribution) onto the constraint set $\mathcal{M}$ (cf. Lemma 3.1 in Csiszár and Shields, 2004). It follows that the arm-size-weighted KL divergence between the unconstrained MLE and the constrained MLE is bounded above by the corresponding divergence between the unconstrained MLE and the true M.

In the following results, I omit the $\mathbf{n}$ subscript from MLEs for legibility, e.g. $\widehat{M}^c := \widehat{M}^c_{\mathbf{n}}$.

Lemma C.3 (CMLE as reverse-KL projection)**.**

$$\widehat{M}^c \in \arg\min_{m \in \mathcal{M}} \sum_{k=1}^{K} n_k D_{\text{KL}}\left(\widehat{M}^u_{\bullet,k} \middle\| m_{\bullet,k}\right),$$

and so for the true M,

$$\sum_{k=1}^{K} n_k D_{\text{KL}}\left(\widehat{M}^u_{\bullet,k} \middle\| \widehat{M}^c_{\bullet,k}\right) \leqslant \sum_{k=1}^{K} n_k D_{\text{KL}}\left(\widehat{M}^u_{\bullet,k} \middle\| M_{\bullet,k}\right).$$

Proof. Rewrite the objective in (7) as

$$\sum_{k=1}^{K} n_k \sum_{y\in\mathcal{Y}} \frac{X_{y,k}}{n_k} \log m_{y,k} = \sum_{k=1}^{K} n_k \sum_{y\in\mathcal{Y}} \widehat{M}^u_{y,k} \log m_{y,k}.$$

Subtracting the constant term $\sum_{k=1}^{K} n_k \sum_{y\in\mathcal{Y}} \widehat{M}^u_{y,k} \log \widehat{M}^u_{y,k}$ from the right-hand side, we get

$$\begin{aligned}\sum_{k=1}^{K} n_k \sum_{y:\widehat{M}^u_{y,k}>0} \widehat{M}^u_{y,k} \log\left(\frac{m_{y,k}}{\widehat{M}^u_{y,k}}\right) &= -\sum_{k=1}^{K} n_k \sum_{y:\widehat{M}^u_{y,k}>0} \widehat{M}^u_{y,k} \log\left(\frac{\widehat{M}^u_{y,k}}{m_{y,k}}\right) \\ &= -\sum_{k=1}^{K} n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k} \middle\| m_{\bullet,k}\right).\end{aligned}$$

□

Lemma C.4.

$$\sup_{M\in\mathcal{M}} \mathbb{E}_M\left[n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k} \middle\| M_{\bullet,k}\right)\right] \leqslant 3.$$

Proof. Let p be absolutely continuous with respect to q. Since $\log u \leqslant u-1$ for any $u>0$, and terms with $p_i = 0$ contribute zero to the sum,

$$\begin{aligned}D_{\mathrm{KL}}(p\|q) &= \sum_{\operatorname{supp} q} p_i \log\left(\frac{p_i}{q_i}\right) \\ &\leqslant \sum_{\operatorname{supp} q} p_i \left(\frac{p_i}{q_i} - 1\right) \\ &= \sum_{\operatorname{supp} q} \left(\frac{p_i^2}{q_i} - p_i + q_i - p_i\right) \\ &= \sum_{\operatorname{supp} q} \frac{(p_i - q_i)^2}{q_i},\end{aligned}$$

where the second equality uses $\sum p_i = 1 = \sum q_i$. Note that $\widehat{M}^u_{\bullet,k}$ is absolutely continuous with respect to $M_{\bullet,k}$ $\mathbb{P}_{M,\mathbf{n}}$-a.s. Then

$$D_{\mathrm{KL}}(\widehat{M}^u_{\bullet,k}\|M_{\bullet,k}) \leqslant \sum_{y:M_{y,k}>0} \frac{(\widehat{M}^u_{y,k} - M_{y,k})^2}{M_{y,k}},$$

and so

$$\begin{aligned}
\mathbb{E}_M\left[n_k D_{\mathrm{KL}}(\widehat{M}^u_{\bullet,k}\|M_{\bullet,k})\right] &\leqslant n_k \sum_{y:M_{y,k}>0} \frac{\mathbb{E}_M\left[(\widehat{M}^u_{y,k}-M_{y,k})^2\right]}{M_{y,k}} \\
&= n_k \sum_{y:M_{y,k}>0} \frac{\mathrm{Var}_M\left[\widehat{M}^u_{y,k}\right]}{M_{y,k}} \\
&= n_k \sum_{y:M_{y,k}>0} \frac{M_{y,k}(1-M_{y,k})/n_k}{M_{y,k}} \\
&= \sum_{y:M_{y,k}>0} (1-M_{y,k}) \\
&\leqslant 3,
\end{aligned}$$

where the second equality uses the variance of $\frac{1}{n_k}\mathrm{Binom}(n_k, M_{y,k})$. □

Proof of Proposition 1. For any two discrete distributions, p and q, denote by $\mathrm{TV}(p,q)$ the total variation distance between them, given by $\frac{1}{2}\sum_i |p_i - q_i|$. Note that no constrained MLE would have $\widehat{M}^c_{y,k} = 0$ if $X_{y,k} > 0$, because then the log likelihood would be $-\infty$, whereas the feasible matrix with every column uniform has finite log likelihood. It follows that each $\widehat{M}^u_{\bullet,k} \ll \widehat{M}^c_{\bullet,k}$. For $k = 1, \ldots, K$, Pinsker's inequality (Boucheron et al., 2013, Theorem 4.19) implies that $\mathbb{P}_{M,\mathbf{n}}$-a.s.,

$$\|\widehat{M}^u_{\bullet,k} - \widehat{M}^c_{\bullet,k}\|_1^2 = 4 \cdot \mathrm{TV}\left(\widehat{M}^u_{\bullet,k}, \widehat{M}^c_{\bullet,k}\right)^2 \leqslant 2 \cdot D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k}\middle\|\widehat{M}^c_{\bullet,k}\right).$$

Thus,

$$\begin{aligned}
\sum_{k=1}^K n_k \|\widehat{M}^u_{\bullet,k} - \widehat{M}^c_{\bullet,k}\|_1^2 &\leqslant 2\sum_{k=1}^K n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k}\middle\|\widehat{M}^c_{\bullet,k}\right) \\
&\leqslant 2\sum_{k=1}^K n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k}\middle\|M_{\bullet,k}\right),
\end{aligned}$$

where the second inequality uses Lemma C.3. Cauchy-Schwarz gives

$$\left(\sum_{k=1}^K \|\widehat{M}^u_{\bullet,k} - \widehat{M}^c_{\bullet,k}\|_1\right)^2 \leqslant \left(\sum_{k=1}^K \frac{1}{n_k}\right)\left(\sum_{k=1}^K n_k \|\widehat{M}^u_{\bullet,k} - \widehat{M}^c_{\bullet,k}\|_1^2\right),$$

so that

$$\left(\sum_{k=1}^K \|\widehat{M}^u_{\bullet,k} - \widehat{M}^c_{\bullet,k}\|_1\right)^2 \leqslant 2\left(\sum_{k=1}^K \frac{1}{n_k}\right)\left(\sum_{k=1}^K n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k}\middle\|M_{\bullet,k}\right)\right).$$

The left-hand side is just $\|\widehat{M}^u - \widehat{M}^c\|_1^2$, so we can write

$$\left\|\widehat{M}^u - \widehat{M}^c\right\|_1 \leqslant \sqrt{2\left(\sum_{k=1}^K \frac{1}{n_k}\right)\left(\sum_{k=1}^K n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k}\middle\|M_{\bullet,k}\right)\right)}.$$

Taking expectations and applying Jensen's inequality to the square root,

$$\mathbb{E}_M\left\|\widehat{M}^u - \widehat{M}^c\right\|_1 \leqslant \sqrt{2\left(\sum_{k=1}^K \frac{1}{n_k}\right)\mathbb{E}_M\left[\sum_{k=1}^K n_k D_{\mathrm{KL}}\left(\widehat{M}^u_{\bullet,k}\middle\|M_{\bullet,k}\right)\right]}.$$

Then Lemma C.4 implies

$$\mathbb{E}_M\left\|\widehat{M}^u - \widehat{M}^c\right\|_1 \leqslant \sqrt{6K\sum_{k=1}^K \frac{1}{n_k}} \leqslant K\sqrt{\frac{6}{n_{\min}}}.$$

We now bound $\sup_M \mathbb{E}_M\|\widehat{M}^u - M\|_1$. For $k = 1, \ldots, K$, by equivalence of ℓ^p-norms,

$$\begin{aligned}
\mathbb{E}_M\left\|\widehat{M}^u_{\bullet,k} - M_{\bullet,k}\right\|_1 &\leqslant \sqrt{4}\cdot\mathbb{E}_M\left\|\widehat{M}^u_{\bullet,k} - M_{\bullet,k}\right\|_2 \\
&= 2\cdot\mathbb{E}_M\left[\sqrt{\sum_{y\in\mathcal{Y}}\left(\widehat{M}^u_{y,k} - M_{y,k}\right)^2}\right] \\
&\leqslant 2\sqrt{\sum_{y\in\mathcal{Y}}\mathbb{E}_M\left[\left(\widehat{M}^u_{y,k} - M_{y,k}\right)^2\right]} \\
&= 2\sqrt{\sum_{y\in\mathcal{Y}}\mathrm{Var}_M\,\widehat{M}^u_{y,k}}. \qquad (13)
\end{aligned}$$

Recall that $\widehat{M}^u_{y,k} = \frac{1}{n_k}\,\mathrm{Binom}(n_k, M_{y,k})$ and $\sum_{y\in\mathcal{Y}} M_{y,k} = 1$ so

$$\begin{aligned}
\sum_{y\in\mathcal{Y}}\mathrm{Var}_M\,\widehat{M}^u_{y,k} &= \frac{1}{n_k}\sum_{y\in\mathcal{Y}} M_{y,k}(1 - M_{y,k}) \\
&= \frac{1 - \sum_{y\in\mathcal{Y}} M_{y,k}^2}{n_k} \\
&= \frac{1 - (\sum_{y\in\mathcal{Y}}\frac{1}{4})(\sum_{y\in\mathcal{Y}} M_{y,k}^2)}{n_k} \\
&\leqslant \frac{1 - (\sum_{y\in\mathcal{Y}}\frac{1}{2}M_{y,k})^2}{n_k} \\
&= \frac{1 - \frac{1}{4}(\sum_{y\in\mathcal{Y}} M_{y,k})^2}{n_k} \\
&= \frac{3}{4n_k},
\end{aligned}$$

where the inequality uses Cauchy-Schwarz. Plugging this back into (13), we get

$$\mathbb{E}_M \left\| \widehat{M}^u_{\bullet,k} - M_{\bullet,k} \right\|_1 \leqslant \sqrt{\frac{3}{n_k}}.$$

Summing over columns $k = 1, \ldots, K$,

$$\mathbb{E}_M \left\| \widehat{M}^u - M \right\|_1 \leqslant K\sqrt{\frac{3}{n_{\min}}}.$$

By the triangle inequality,

$$\begin{aligned}\mathbb{E}_M \left\| \widehat{M}^c - M \right\|_1 &\leqslant \mathbb{E}_M \left\| \widehat{M}^c - \widehat{M}^u \right\|_1 + \mathbb{E}_M \left\| \widehat{M}^u - M \right\|_1 \\ &\leqslant K\frac{\sqrt{6}+\sqrt{3}}{\sqrt{n_{\min}}}.\end{aligned}$$

Taking the supremum over $M \in \mathcal{M}$ gives the desired result. □

Proof of Proposition 2. Recall from Appendix B that (8) can be written as

$$\begin{aligned}\inf_{a \in \mathcal{A}} \sup_{t \in \{1,\ldots,T\}} \sup_{\mathbf{q}} \quad & \operatorname{vec}(w(e_t - a)^\mathsf{T})^\mathsf{T} S\mathbf{q} - c^\mathsf{T}(e_t - a) \\ \text{s.t.} \quad & \Psi S\mathbf{q} = \mathbf{m} \\ & \mathbf{1}^\mathsf{T}\mathbf{q} = 1 \\ & \mathbf{q} \geqslant 0.\end{aligned}$$

Note that

$$\begin{aligned}\operatorname{vec}(w(e_t - a)^\mathsf{T})^\mathsf{T} S &= \operatorname{vec}(we_t^\mathsf{T})^\mathsf{T} S - \operatorname{vec}(wa^\mathsf{T})^\mathsf{T} S \\ &= \operatorname{vec}(we_t^\mathsf{T})^\mathsf{T} S - \sum_{s=1}^{T} a_s \operatorname{vec}(we_s^\mathsf{T})^\mathsf{T} S \\ &= r_t^\mathsf{T} - a^\mathsf{T} R^\mathsf{T}.\end{aligned}$$

Denoting $A := \Psi S$, write the inner linear program as

$$\begin{aligned}\sup_{\mathbf{q}} \quad & (r_t - Ra)^\mathsf{T}\mathbf{q} - c^\mathsf{T}(e_t - a) \\ \text{s.t.} \quad & A\mathbf{q} = \mathbf{m} \\ & \mathbf{1}^\mathsf{T}\mathbf{q} = 1 \\ & \mathbf{q} \geqslant 0.\end{aligned}$$

Since $\mathbf{m} \in \mathcal{M}$, this linear program is feasible; since the constraint set is a subset of the simplex, the linear program is also bounded. By the Duality Theorem (Luenberger and Ye,

2008, p83), the dual program also has an optimal solution, and the optimal value coincides with that of the primal. The cost term does not depend on $\mathbf{q}$, so the dual program is

$$\begin{aligned} \inf_{u_t,\lambda_t} \quad & \mathbf{m}^\mathsf{T} u_t + \lambda_t - c^\mathsf{T}(e_t - a) \\ \text{s.t.} \quad & A^\mathsf{T} u_t + \lambda_t \mathbf{1} + Ra \geqslant r_t. \end{aligned}$$

In epigraph form (Boyd and Vandenberghe, 2004, p134), this is

$$\begin{aligned} \inf_{z,u_t,\lambda_t} \quad & z \\ \text{s.t.} \quad & z \geqslant \mathbf{m}^\mathsf{T} u_t + \lambda_t - c^\mathsf{T}(e_t - a) \\ & A^\mathsf{T} u_t + \lambda_t \mathbf{1} + Ra \geqslant r_t. \end{aligned}$$

The profiled loss is the envelope over these programs for $t = 1, \ldots, T$:

$$\begin{aligned} \sup_{t=1,\ldots,T} \inf_{z,u_t,\lambda_t} \quad & z \\ \text{s.t.} \quad & z \geqslant \mathbf{m}^\mathsf{T} u_t + \lambda_t - c^\mathsf{T}(e_t - a) \\ & A^\mathsf{T} u_t + \lambda_t \mathbf{1} + Ra \geqslant r_t, \end{aligned}$$

which can be written

$$\begin{aligned} \inf_{z,(u_t,\lambda_t)_{t=1}^T} \quad & z \\ \text{s.t.} \quad & z \geqslant \mathbf{m}^\mathsf{T} u_t + \lambda_t - c^\mathsf{T}(e_t - a) && t = 1, \ldots, T \\ & A^\mathsf{T} u_t + \lambda_t \mathbf{1} + Ra \geqslant r_t && t = 1, \ldots, T. \end{aligned}$$

Minimising over $a \in \mathcal{A}$,

$$\begin{aligned} \inf_{a,z,(u_t,\lambda_t)_{t=1}^T} \quad & z \\ \text{s.t.} \quad & z \geqslant \mathbf{m}^\mathsf{T} u_t + \lambda_t - c^\mathsf{T}(e_t - a) && t = 1, \ldots, T \\ & A^\mathsf{T} u_t + \lambda_t \mathbf{1} + Ra \geqslant r_t && t = 1, \ldots, T \\ & \mathbf{1}^\mathsf{T} a = 1 \\ & a \geqslant 0. \end{aligned}$$

□

D Multiple Ordered Endpoints

This appendix generalises the model from two binary endpoints to any finite number of endpoints, each with a finite number of ordered values. For each individual, a binary endpoint is described by one threshold dose; an endpoint with several levels is described by

one threshold dose for each level above zero. The main cost is computational. The dimension of the plug-in linear program grows combinatorially with the number and dimension of outcomes.

Unless otherwise stated, $\leqslant$ is the canonical product order on Euclidean space, i.e.

$$x \leqslant x' \iff x_i \leqslant x'_i \text{ for all } i.$$

Outcomes. Suppose that there are $E \in \mathbb{N}$ endpoints, with respective dose-response functions

$$y_e^j : \{1, \ldots, T\} \to \{0, 1, \ldots, \overline{y}_e\},$$

where $\overline{y}_e \geqslant 1$. I will often suppress the individual-indexing superscript j. Define the outcome-vector support

$$\mathcal{Y} := \bigtimes_{e=1}^{E} \{0, \ldots, \overline{y}_e\}$$

with cardinality

$$d_{\mathcal{Y}} := \#\mathcal{Y} = \prod_{e=1}^{E} (\overline{y}_e + 1).$$

For matrices and vectors, order outcomes lexicographically.

The monotone-dose-response assumption is that

$$t' \geqslant t \implies y^j(t') \geqslant y^j(t), \qquad t, t' \in \{1, \ldots, T\},\ j \in \mathcal{J}.$$

Collect the marginal potential-outcome distributions under $t = 1, \ldots, T$ in a $d_{\mathcal{Y}} \times T$ column-stochastic matrix P with entries

$$P_{y,t} := \mathbb{P}(y(t) = y), \qquad y \in \mathcal{Y}, \quad t = 1, \ldots, T.$$

Threshold representation. A monotone dose-response function can be written using $\overline{y}_e$ threshold doses:

$$y_e(t) = \sum_{h=1}^{\overline{y}_e} \mathbf{1}\{t \geqslant \theta_{e,h}\}.$$

Denote the threshold vector for endpoint e by

$$\boldsymbol{\theta}_e := (\theta_{e,1}, \ldots, \theta_{e,\overline{y}_e}) \in \Theta_e,$$

where

$$\Theta_e := \left\{ \boldsymbol{\theta}_e \in \{1, \ldots, T+1\}^{\overline{y}_e} \,\middle|\, \theta_{e,1} \leqslant \cdots \leqslant \theta_{e,\overline{y}_e} \right\}.$$

Equal thresholds are allowed and encode jumps by more than one level at a dose. If $\theta_{e,h} = 1$ then all doses give $y_e(t) \geqslant h$; if $\theta_{e,h} = T + 1$ then all doses give $y_e(t) < h$. Collect the threshold vectors $\boldsymbol{\theta}_e$ across endpoints e in a threshold profile

$$\boldsymbol{\theta} := (\boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_E)$$

and define

$$\Theta := \bigtimes_{e=1}^{E} \Theta_e,$$

which has cardinality[10]

$$d_\Theta := \#\Theta = \prod_{e=1}^{E} \binom{T + \overline{y}_e}{T}.$$

A population distribution over monotone dose-response functions y is equivalent to a distribution over threshold profiles Θ. Denote such a threshold distribution by $\mathbf{q} \in \Delta^{d_\Theta - 1}$, where $q_{\boldsymbol{\theta}}$ is the probability assigned to profile $\boldsymbol{\theta} \in \Theta$.

Order threshold profiles lexicographically, and use this order for the entries of $\mathbf{q}$ and the columns of $s(t)$.[11] For $\boldsymbol{\theta} \in \Theta$, define

$$y(t, \boldsymbol{\theta}) := \left(\sum_{h=1}^{\overline{y}_1} \mathbf{1}\{t \geqslant \theta_{1,h}\}, \ldots, \sum_{h=1}^{\overline{y}_E} \mathbf{1}\{t \geqslant \theta_{E,h}\} \right) \in \mathcal{Y}.$$

For each dose t, define the $d_\mathcal{Y} \times d_\Theta$ binary matrix $s(t)$ by

$$s(t)_{y,\boldsymbol{\theta}} := \mathbf{1}\{y(t, \boldsymbol{\theta}) = y\}, \qquad y \in \mathcal{Y}, \quad \boldsymbol{\theta} \in \Theta.$$

Then $P_{\bullet,t} = s(t)\mathbf{q}$ and stacking gives

$$\mathbf{p} := \operatorname{vec} P = S\mathbf{q}, \quad S := \begin{bmatrix} s(1) \\ s(2) \\ \vdots \\ s(T) \end{bmatrix}.$$

Parameter space and identified subparameter. Let $\mathcal{P}$ be the set of all $d_\mathcal{Y} \times T$ column-stochastic matrices P that are consistent with the monotone dose-response assumption, or equivalently, that can be generated as $\mathbf{p} = S\mathbf{q}$.

A set $L \subseteq \mathcal{Y}$ is a lower set of $\mathcal{Y}$ if $y \in L$ and $y' \leqslant y$ imply $y' \in L$. Let $\mathcal{L}(\mathcal{Y})$ be the collection of nonempty proper lower sets.

Lemma D.1 (Polyhedral characterisation of $\mathcal{P}$)**.** *$P \in \mathcal{P}$ if and only if P is a $d_\mathcal{Y} \times T$ column-stochastic matrix that satisfies the following inequality, for $t = 1, \ldots, T-1$ and for all $L \in \mathcal{L}(\mathcal{Y})$:*

$$\sum_{y \in L} P_{y,t} \geqslant \sum_{y \in L} P_{y,t+1}.$$

Equivalently, $P_{\bullet,t+1}$ first-order stochastically dominates $P_{\bullet,t}$ on $\mathcal{Y}$.

[10] This can be shown using the Stars-and-Bars Theorem.

[11] For example, with two ternary endpoints and $T = 2$, each profile takes the form $(\theta_{1,1}, \theta_{1,2}, \theta_{2,1}, \theta_{2,2})$. The order begins $(1,1,1,1)$, $(1,1,1,2)$, $(1,1,1,3)$, $(1,1,2,2)$, ….

Proof. The proof is analogous to that for the 2×2 case (Lemma 1). The inequalities are exactly those required for stochastic dominance on $\mathcal{Y}$. Necessity follows from monotone dose response. For sufficiency, apply Strassen's theorem and the construction in Appendix C, with $\mathcal{Y}$ in place of $\{0,1\}^2$. □

Let M be the column submatrix consisting of only those columns $t_1, \ldots, t_K$ corresponding to the tested doses. Let C be the $T \times K$ binary tested-dose-selector matrix defined in Appendix B. Then

$$M = PC,$$

and

$$\mathbf{m} := \operatorname{vec} M = \Psi \mathbf{p}, \quad \Psi := C^{\mathsf{T}} \otimes I_{d_{\mathcal{Y}}}.$$

Lemma D.2 (Polyhedral characterisation of $\mathcal{M}$)**.** *$M \in \mathcal{M}$ if and only if M is a $d_{\mathcal{Y}} \times K$ column-stochastic matrix that satisfies the following inequality, for $k = 1, \ldots, K-1$ and for all $L \in \mathcal{L}(\mathcal{Y})$:*

$$\sum_{y \in L} M_{y,k} \geqslant \sum_{y \in L} M_{y,k+1}.$$

Proof. Necessity is immediate from Lemma D.1. Sufficiency can be shown by generating a valid $P \in \mathcal{P}$ from M as in the proof of Lemma 2. □

Data, constrained MLE, and plug-in guarantees. The trial data are now $X \in \mathbb{Z}_{\geqslant 0}^{d_{\mathcal{Y}} \times K}$ with independent columns

$$X_{\bullet,k} \sim \text{Multinom}(n_k, M_{\bullet,k}).$$

A constrained MLE of M is any measurable selection

$$\widehat{M}^c_{\mathbf{n}}(X) \in \arg\max_{m \in \mathcal{M}} \sum_{k=1}^{K} \sum_{y \in \mathcal{Y}} X_{y,k} \log m_{y,k}.$$

Proposition D.1 (Uniform L^1 bound for constrained MLE)**.**

$$\sup_{M \in \mathcal{M}} \mathbb{E}_{M,\mathbf{n}} \left\| \widehat{M}^c_{\mathbf{n}} - M \right\|_1 \leqslant K \frac{\sqrt{2(d_{\mathcal{Y}} - 1)} + \sqrt{d_{\mathcal{Y}} - 1}}{\sqrt{n_{\min}}}.$$

Proof. Drop the $\mathbf{n}$ subscripts for legibility. The weighted KL-projection inequality in Lemma C.3 still holds. Following the proof of Lemma C.4,

$$\mathbb{E}_M \left[n_k D_{\mathrm{KL}} \left(\widehat{M}^u_{\bullet,k} \middle\| M_{\bullet,k} \right) \right] \leqslant \sum_{y : M_{y,k} > 0} (1 - M_{y,k}) \leqslant d_{\mathcal{Y}} - 1.$$

As in the proof of Proposition 1, Pinsker's inequality and the variance calculation then give

$$\mathbb{E}_M \left\| \widehat{M}^u - \widehat{M}^c \right\|_1 \leqslant K \sqrt{\frac{2(d_{\mathcal{Y}} - 1)}{n_{\min}}},$$

$$\mathbb{E}_M \left\| \widehat{M}^u - M \right\|_1 \leqslant K \sqrt{\frac{d_{\mathcal{Y}} - 1}{n_{\min}}}.$$

Combining these bounds using the triangle inequality gives the result. □

Fix some d_Y-vector of welfare values w.

Proposition D.2 (Plug-in guarantees and computation)**.** *With the outcome space defined in this appendix, Theorem 1 and Proposition 2 continue to hold. In particular, for the plug-in rule using the constrained MLE $\widehat{M}^c_{\mathbf{n}}$ and the ℓ^1 norm on $\mathcal{M}$,*

$$\varepsilon^c_{\mathbf{n}} \leqslant 2C_1 K \frac{\sqrt{2(d_Y - 1)} + \sqrt{d_Y - 1}}{\sqrt{n_{\min}}} = O\left(n_{\min}^{-1/2}\right),$$

where C_1 is the Lipschitz constant for L^{prof} in the ℓ^1 norm on $\mathcal{M}$. In the linear program (9), use S and Ψ defined in this appendix, $u_t \in \mathbb{R}^{d_Y K}$, $r_t := S^\mathsf{T} \operatorname{vec}(w e_t^\mathsf{T}) \in \mathbb{R}^{d_\Theta}$, and $R \in \mathbb{R}^{d_\Theta \times T}$ with columns $r_1, \ldots, r_T$.

Proof. The proofs of Lemma 5, Theorem 1 and Proposition 2 are unchanged, up to the dimensions of the objects. Combining Theorem 1 with Proposition D.1 gives the CMLE bound. □

Computational challenges. The plug-in linear program has decision-variable dimension

$$2T + 1 + TKd_Y = 2T + 1 + TK \prod_{e=1}^{E} (\overline{y}_e + 1)$$

and number of constraints

$$2T + 1 + Td_\Theta = 2T + 1 + T \prod_{e=1}^{E} \binom{T + \overline{y}_e}{T}.$$

An additional consideration is the number of outcome cells in the experimental data relative to the smallest trial-arm sample size: for desirable statistical properties we would ideally have $n_{\min} \gg d_Y$.

4×2 model example. Consider the model in which the efficacy endpoint takes four ordered values and the adverse-effects endpoint is binary, as in the empirical illustration. Then the space of threshold vectors is

$$\Theta = \{(\theta_{11}, \theta_{12}, \theta_{13}, \theta_{21}) \in \{1, \ldots, T+1\}^4 \mid \theta_{11} \leqslant \theta_{12} \leqslant \theta_{13}\},$$

and $\mathbf{q}$ is a distribution thereon. P is the $8 \times T$ matrix with entries $P_{y,t} = \mathbb{P}(y(t) = y)$ and rows indexed by $y \in \{0,1,2,3\} \times \{0,1\}$ in lexicographic order.

The following figure depicts the nontrivial lower sets of $\{0,1,2,3\} \times \{0,1\}$ as shaded collections of cells in a rectangular grid. The 13 different shadings indicate the $13(T-1)$

different monotonicity inequalities that define $\mathcal{P}$. So, for example, the top-left cell matrix shades only outcome $(0, 0)$ and indicates that

$$P_{00,t} \geqslant P_{00,t+1};$$

the cell matrix to its right shades outcomes $(0, 0)$ and $(1, 0)$ and indicates that

$$P_{00,t} + P_{10,t} \geqslant P_{00,t+1} + P_{10,t+1},$$

and so forth.

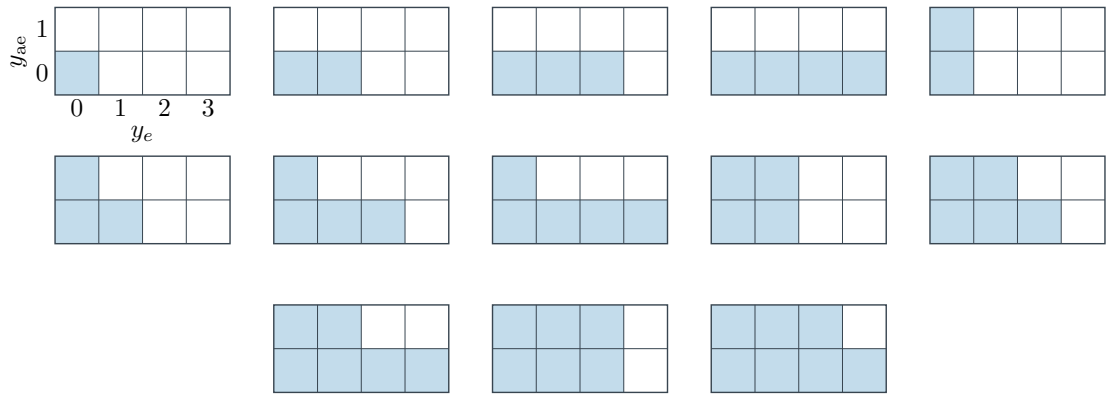


Figure 2: Lower sets in the 4×2 case.

In particular, for computing the constrained MLE analogous to that defined in Section 4.2, the $13(K-1)$ inequalities implied here are the analogues of the $4(K-1)$ monotonicity constraints required in the 2×2 case.

E Empirical Illustration Details

I construct the joint distribution of time to resolution of illness and incidence of nausea or vomiting using the publicly released clinical study reports WV15670 and WV15671 (Jefferson et al., 2014a). These give, for each complete-case participant, the time to alleviation of all symptoms and a list of any adverse effects they experienced. When choosing endpoints, I also considered two later reviews of these (and other) studies on oseltamivir: Dobson et al. (2015) and Jefferson et al. (2014b). Both report decreases in time to alleviation of symptoms and increases in incidence of nausea and vomiting.

I pool the two original studies, assuming a common joint outcome distribution at each tested dose. I omit 156 participants whose records are incomplete: 74 whose records do not establish whether they experienced nausea or vomiting, 3 with missing resolution-time data, and 79 with resolution times that are censored before 168 hours. I retain 65 participants with resolution times that are censored at or after 168 hours and whose outcome can thus be classified into one of my four efficacy levels. Both endpoint values are known for each

Resolution time (hours)	Adverse effects	Placebo	75 mg	150 mg	Total
> 168	No	84	43	64	191
	Yes	10	29	8	47
$(96, 168]$	No	98	73	66	237
	Yes	7	18	25	50
$(48, 96]$	No	103	94	97	294
	Yes	9	20	27	56
≤ 48	No	74	99	94	267
	Yes	3	20	21	44
Total		388	396	402	1,186

Table 4: Participant counts by outcome and treatment dose

participant in the analysis sample. Table 4 gives the number of retained participants in each outcome category at each tested dose.

To calibrate the welfare associated with different outcomes, I use the midpoint of the £25,000 and £35,000 per QALY values used in the National Institute for Health and Care Excellence's (NICE) economic-evaluation guidance (NICE, 2026, Section 4.2.16), and the midpoint of NICE's average decrement in quality of life of 0.22 to 0.23 for otherwise healthy populations experiencing the later phase of an influenza episode (NICE, 2009, Section 4.3.14). Together, these imply a welfare gain of approximately £18.49 per day of earlier illness resolution. I normalise the welfare associated with the pair > 168 hours' resolution time and no adverse effects to £0. For each of the three shorter resolution-time categories, I subtract the median resolution time among retained participants with observed resolution times in that category from the corresponding median for the > 168 category. I multiply this by the hourly value $£30{,}000 \cdot 0.225/(365 \cdot 24)$ to obtain the welfare w_{e0} of the corresponding resolution-time category when no adverse effects are experienced. Vomiting and nausea induced by oseltamivir typically subside within 1–2 days (Roche Products Limited, 2021, Section 4.8). I use the midpoint of this range and a daily welfare decrement of vomiting and nausea equal to that of influenza illness to get a cumulative baseline adverse-effects welfare decrement of £27.74. I assume that adverse effects have a smaller marginal welfare effect on individuals that recover quickly from their illness, setting $w_{e1} = 1.05w_{e0} - 27.74$. The resulting welfare values are listed in Table 2.[12] For the right panel of Figure 1, let $m \in [0, 2]$ denote the multiplier on the horizontal axis. I set

$$w_{e1}(m) = w_{e0} - m\,(27.74 - 0.05w_{e0})\,,$$

[12]Note that the statistical guarantees in Section 4 do not account for excluding participants with incomplete outcomes or calibrating welfare from the same data.

with $m = 1$ corresponding to the baseline.

For the cost of each dose, I use the cheapest exact combination of 30, 45, and 75 mg capsules at the NHS list prices (UKHSA, 2026, Table 9). I multiply these October 2025 prices by the CPI ratio 144.1/139.8 to express them in September 2026 pounds (Office for National Statistics, 2026). The full-course costs of each of the ten doses, rounded for display, are then

$$c = (0, 7.95, \underbrace{15.88, 15.89, 15.88}_{\text{45–75 mg}}, \underbrace{23.84, 23.83}_{\text{90–105 mg}}, \underbrace{31.77, 31.78, 31.77}_{\text{120–150 mg}}).$$

Notably, certain adjacent doses have equal or approximately equal cost. Figure 3 instead shows the corresponding plug-in mixtures with linear cost function $c(t) = 7.71(144.1/139.8)d_t/30$, where d_t is the dose in mg corresponding to index t. Over this range of cost multipliers, more doses receive positive probability when costs are linear.

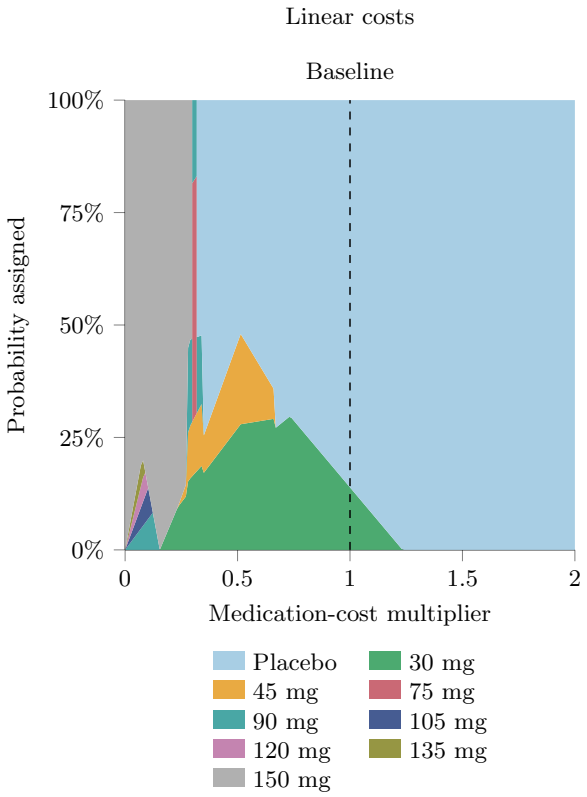


Figure 3: Plug-in dose allocations at different medication costs when costs are linear. A multiplier of one denotes the baseline linear cost function. All other inputs are held fixed.